

Chapter 14: Tensor Visualization

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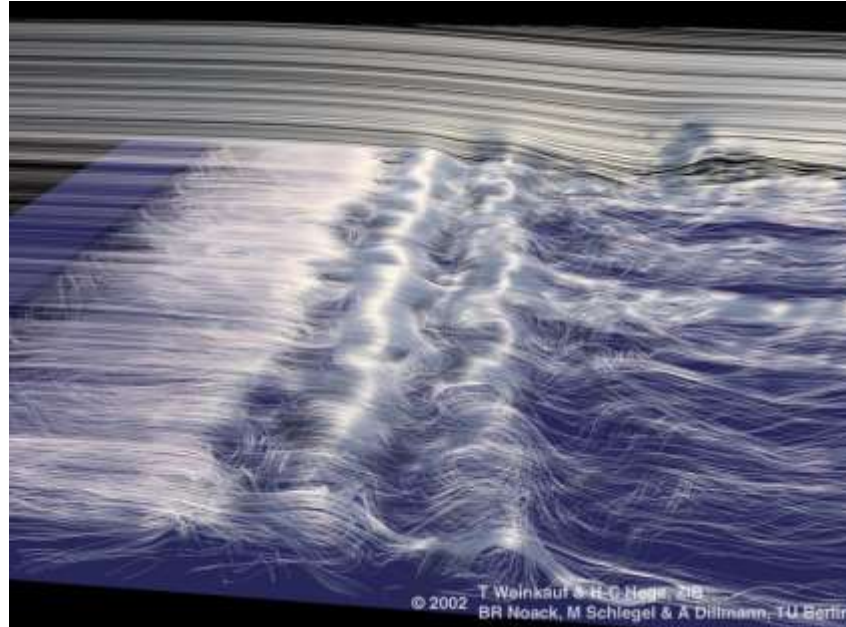
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July 7/8, 2026

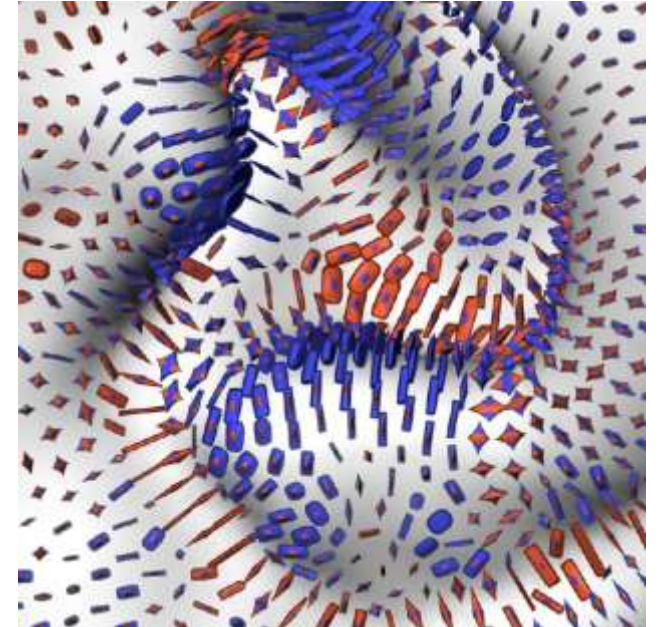
Where We Are In Scientific Visualization



Scalar Fields



Vector Fields



Tensor Fields

$$f: D \rightarrow R^{m \times m} \quad \text{with} \quad D \subseteq R^n$$

Section 14.1:

Introduction to Tensor Fields

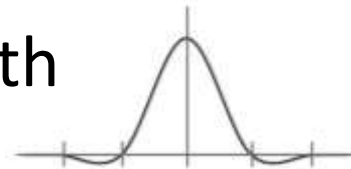
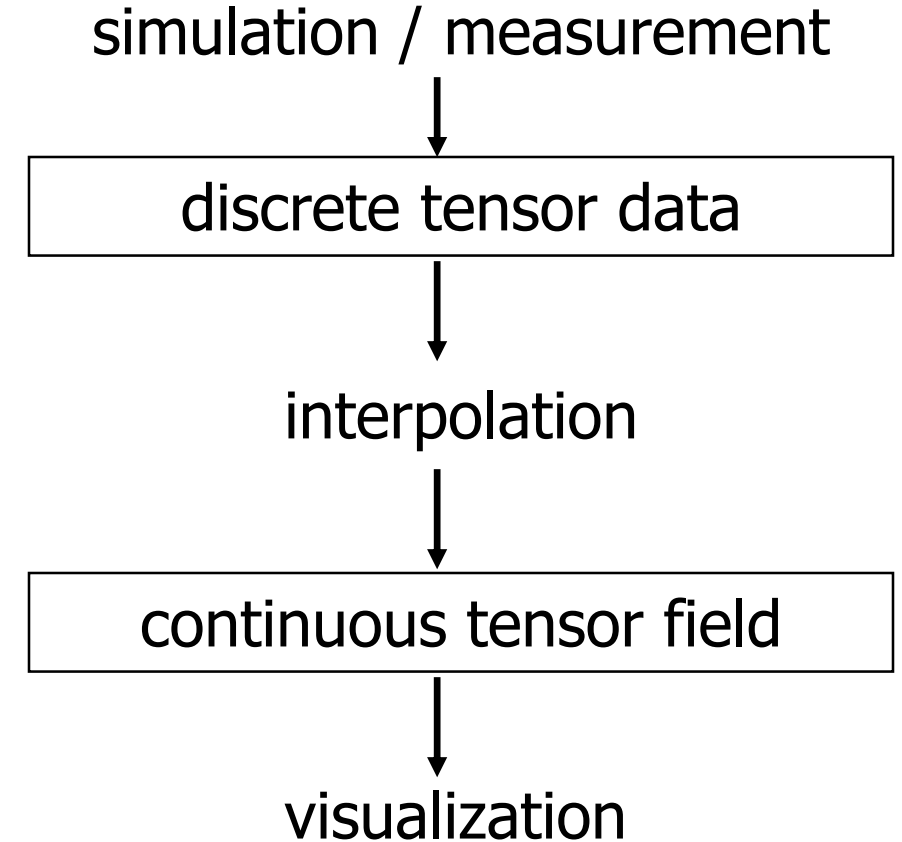
Classification of Tensor Fields

We will focus on

- **matrices**, which we refer to as second-order tensors
 - terminology based on tensor product (matrices as linear maps)
 - in other contexts (e.g., deep learning), “tensor” = multi-way array
- square matrices, which can be...
 - **symmetric** or **unsymmetric**
 - we will mostly focus on the symmetric case
 - **(semi-)definite** or **indefinite**
 - Positive semi-definite: $\mathbf{v}^T \mathbf{M} \mathbf{v} \geq 0$ for all $\mathbf{v}$
 - Easily seen from non-negative eigenvalues of $\mathbf{M}$
 - Strictly positive definite: $\mathbf{v}^T \mathbf{M} \mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$

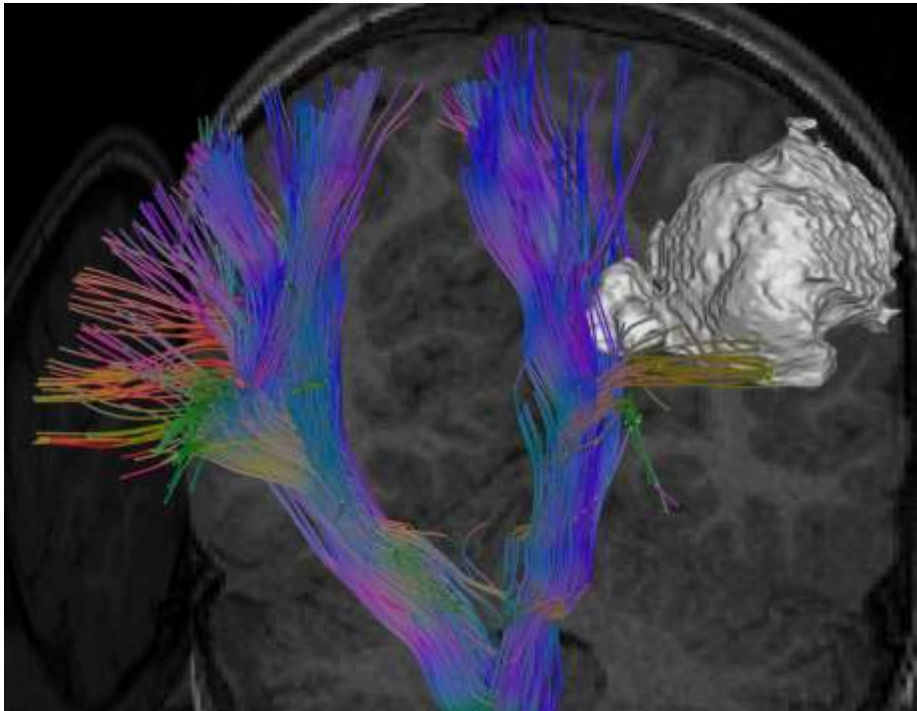
Tensor Field Interpolation

- Similar to vector data, **tensor fields** can be reconstructed from discrete samples by **component-wise interpolation**
- **(Semi-)definiteness** is preserved if interpolation takes convex combinations
 - *Examples:* bi-/trilinear interpolation, scattered data interpolation with inverse distance weights
 - *Counter-example:* Interpolation with Catmull-Rom splines (8.4)

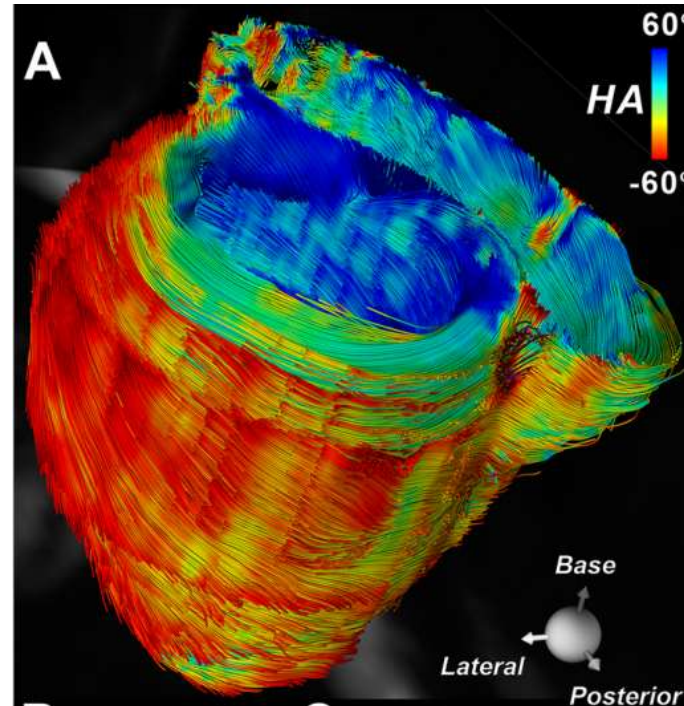


Example: Diffusion Tensor Imaging

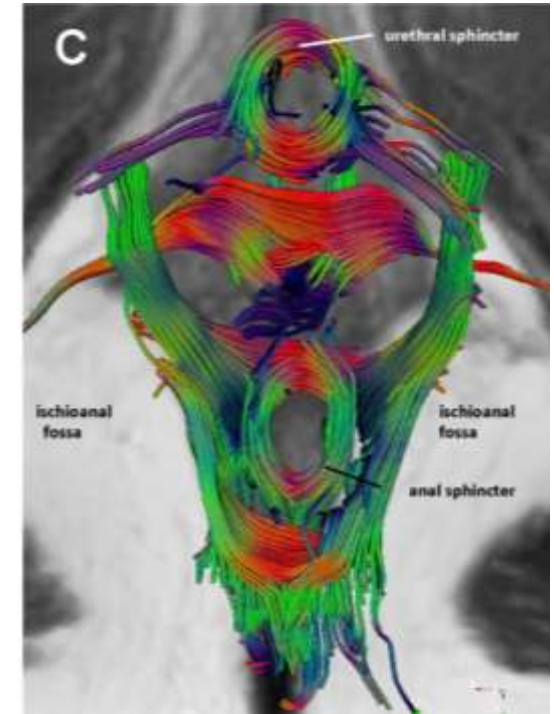
- Medical imaging modality that can visualize fibrous tissue (e.g., white matter tracts or muscles)
 - Second-order, symmetric, positive definite



[Gruen et al., NeuroImage 2023]



[Mekkaoui et al. 2015]



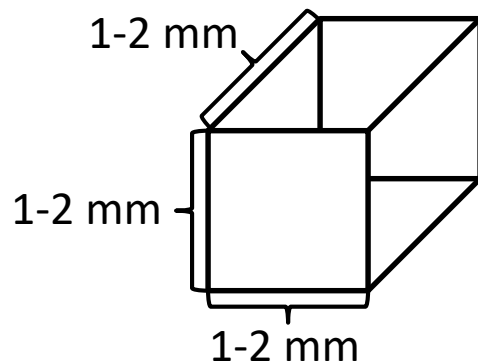
[Zijta et al. 2012]

Introduction to Diffusion MRI

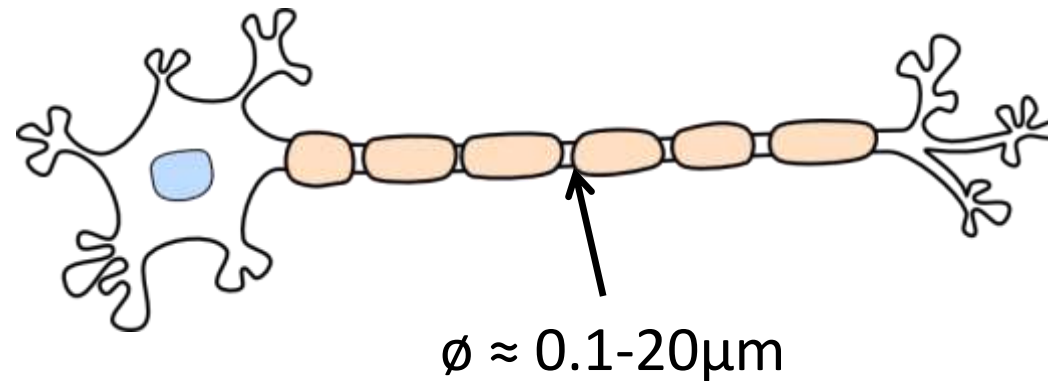
Goal: Investigate the microstructure of biological tissue using Magnetic Resonance Imaging (MRI)



Challenge: Voxel size is far too large to resolve the structures of interest



Voxel size



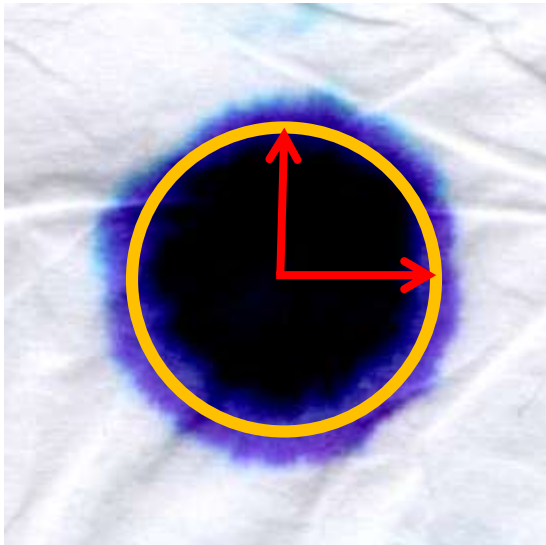
Axon (nerve fiber) size

Introduction to Diffusion MRI

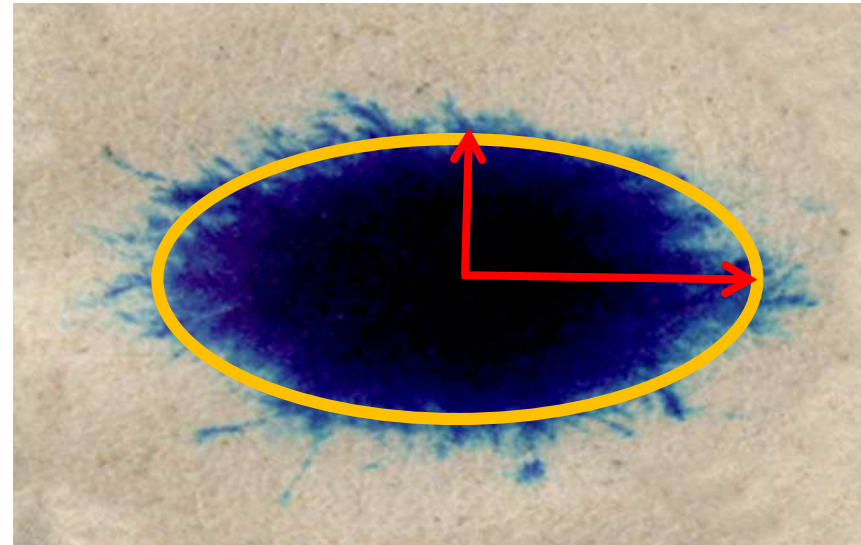
Approach: Use water molecules as a contrast agent

- Exploits their spontaneous heat motion at the desired spatial scale

Analogy: Observe diffusion of ink on paper



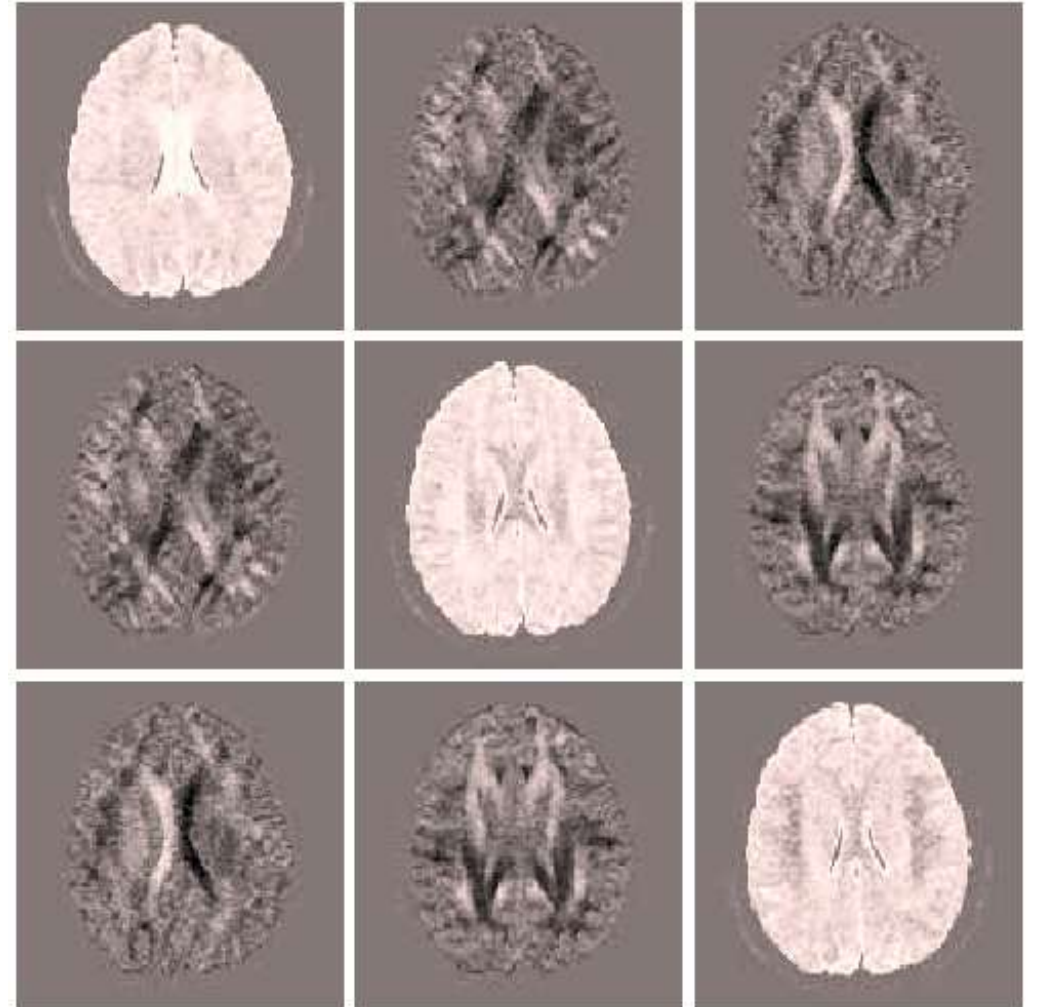
Kleenex



Newspaper

Component-wise Visualization

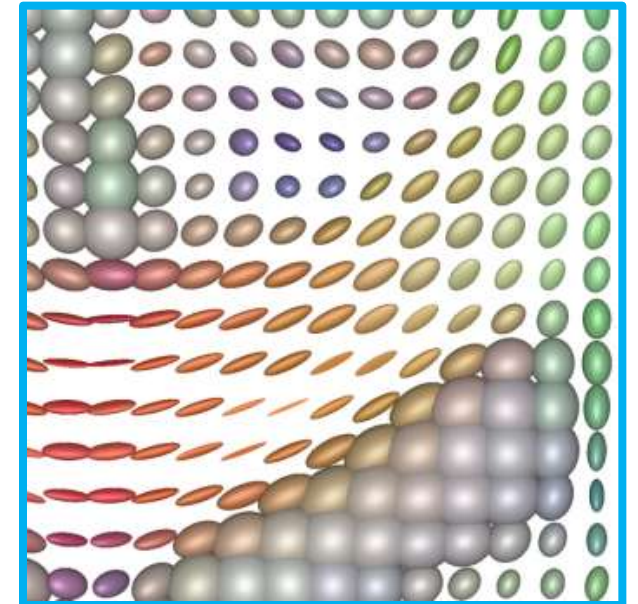
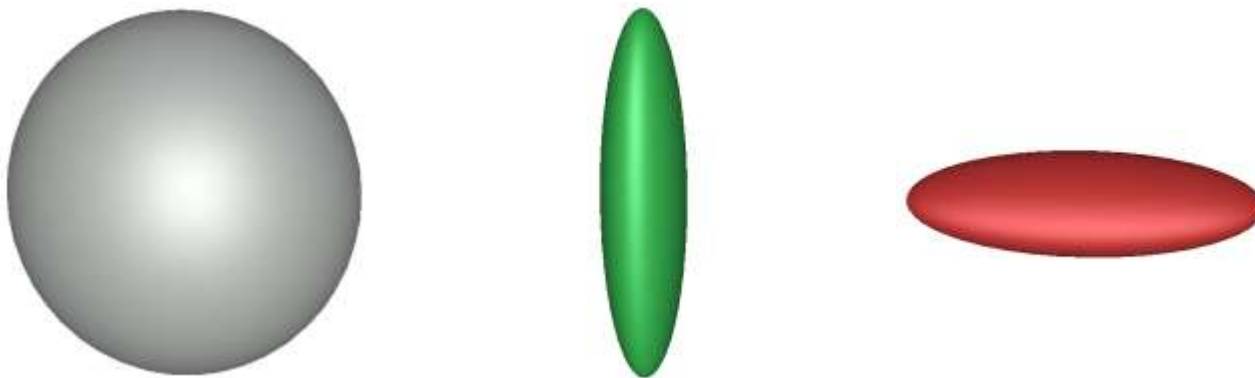
- Showing tensor fields as a matrix of scalar fields includes all information, but in a way that is difficult to interpret
- Motivates the need for the visualization techniques in this chapter



Interpretation of Diffusion Tensors

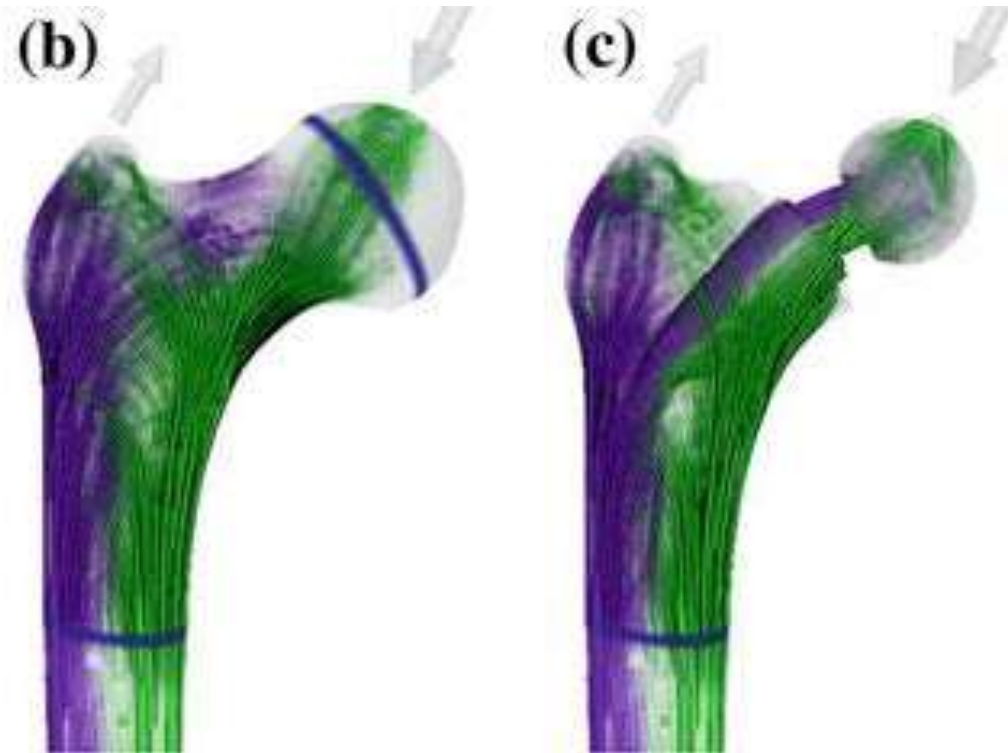
Diffusion tensor **D** can be understood in terms of

- 3 eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$
 - *Note:* Since diffusivity is a non-negative physical quantity, all $\lambda > 0$. Negative λ 's can result from measurement noise and are frequently set to 0 or a small ϵ
- 3 orthogonal eigenvectors $\mathbf{e}_1/\mathbf{e}_2/\mathbf{e}_3$
 - $\mathbf{e}_1$ indicates single main fiber orientation

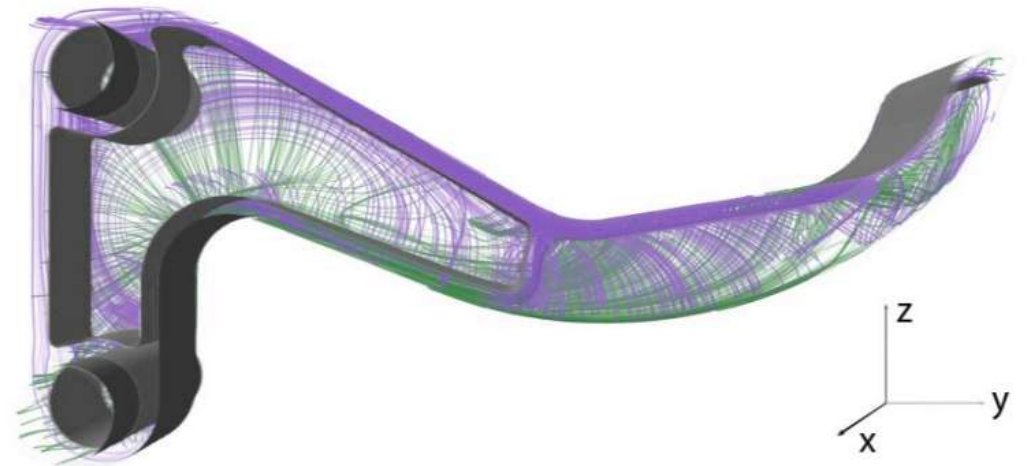


Example: Stress Tensors

- Describes **compressive** (negative eigenvalues) and **tensile** (positive evals) stress on a material
 - Second-order, symmetric, indefinite



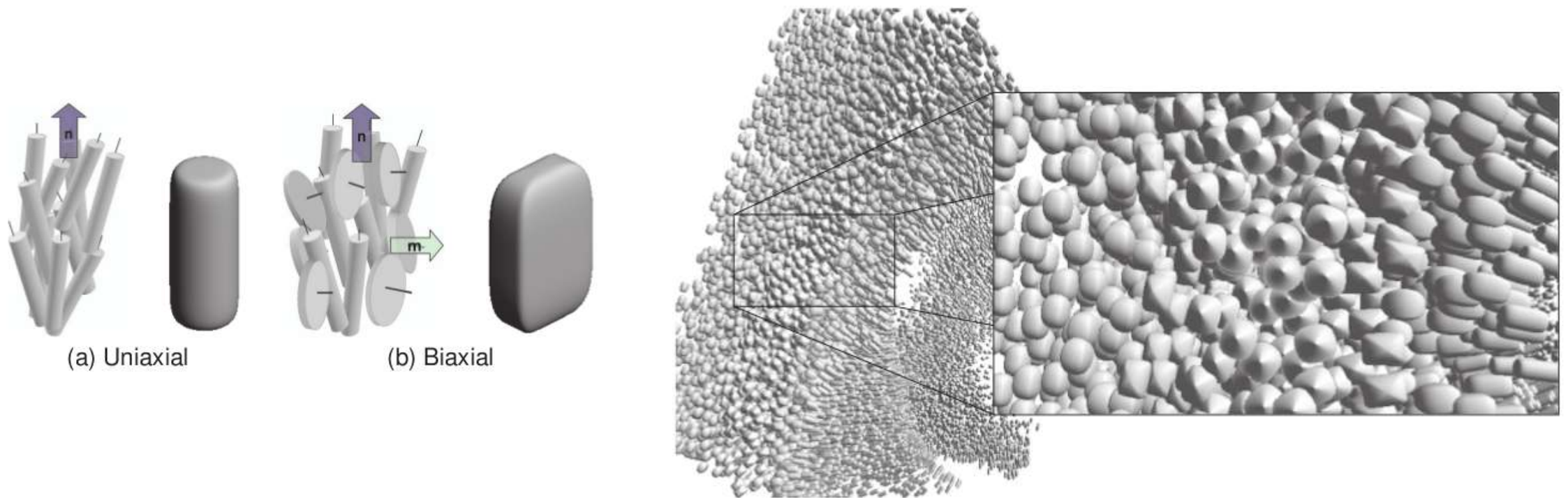
[Yang et al., Comp Vis Sci 2011]



[Kratz et al., PacificVis 2014]

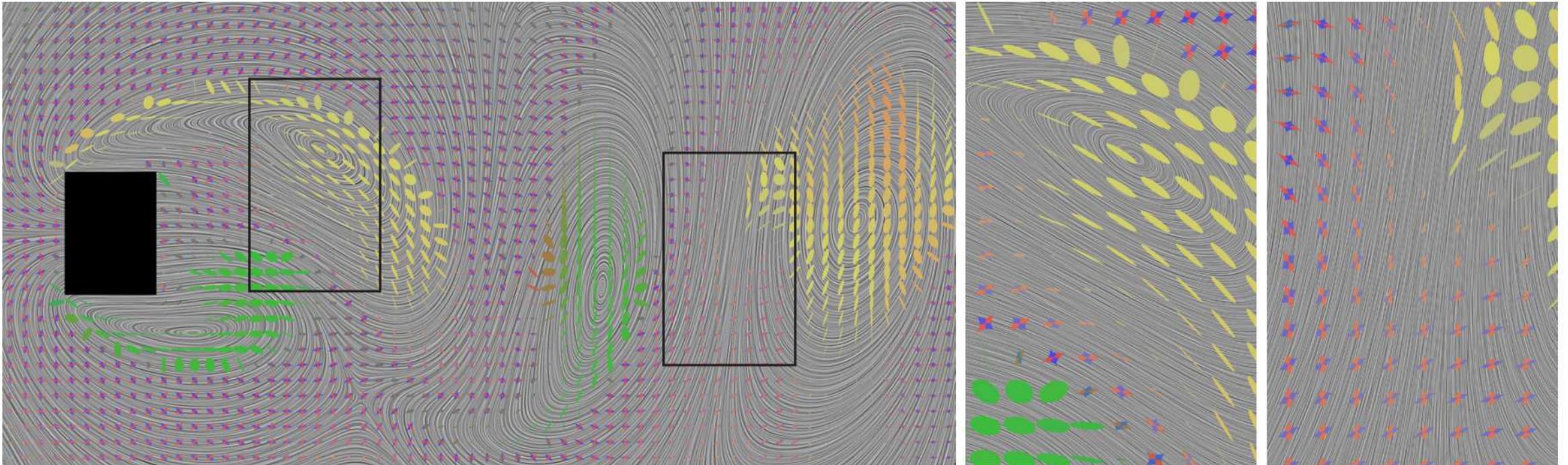
Example: Nematic Liquid Crystal Alignment

- Tensor describes the overall alignment of elongated molecules in nematic liquid crystals
 - Second-order, symmetric, traceless (indefinite)



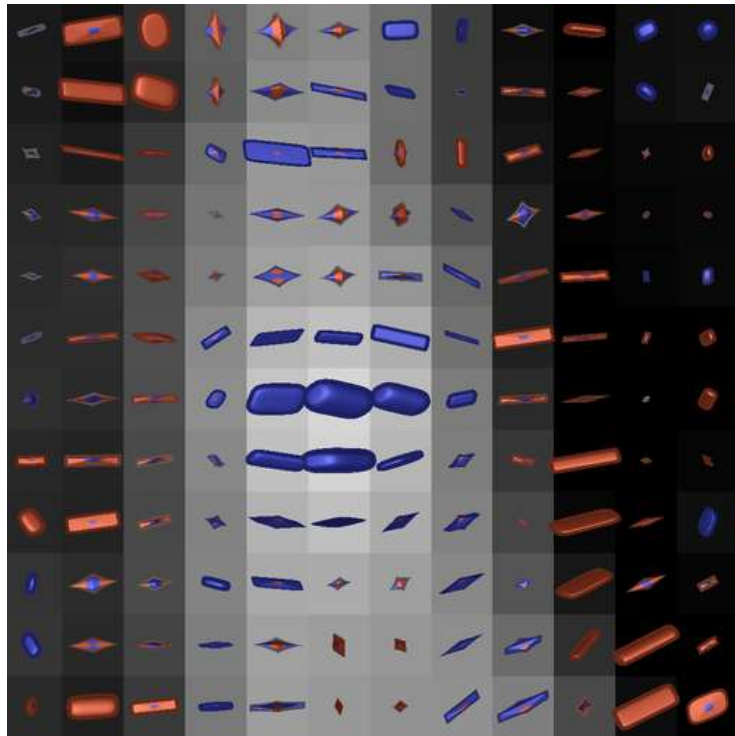
Example: Vector Field Jacobian

- **Reminder:** The Jacobian contains all (first order) partial derivatives of a vector field
 - Second-order, non-symmetric

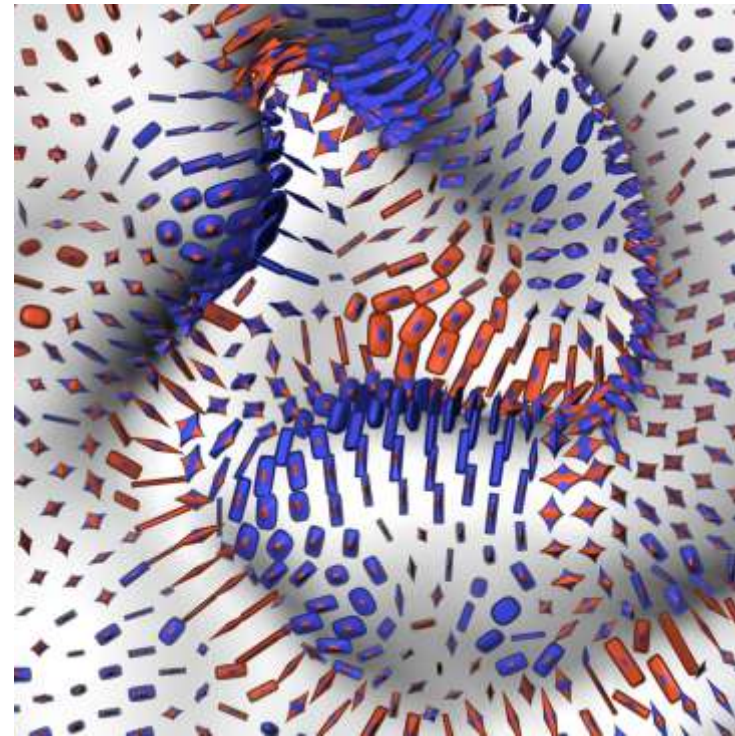


Example: Hessian of Scalar Fields

- **Reminder:** The Hessian matrix contains all second partial derivatives of a scalar field
 - Second-order, symmetric, indefinite



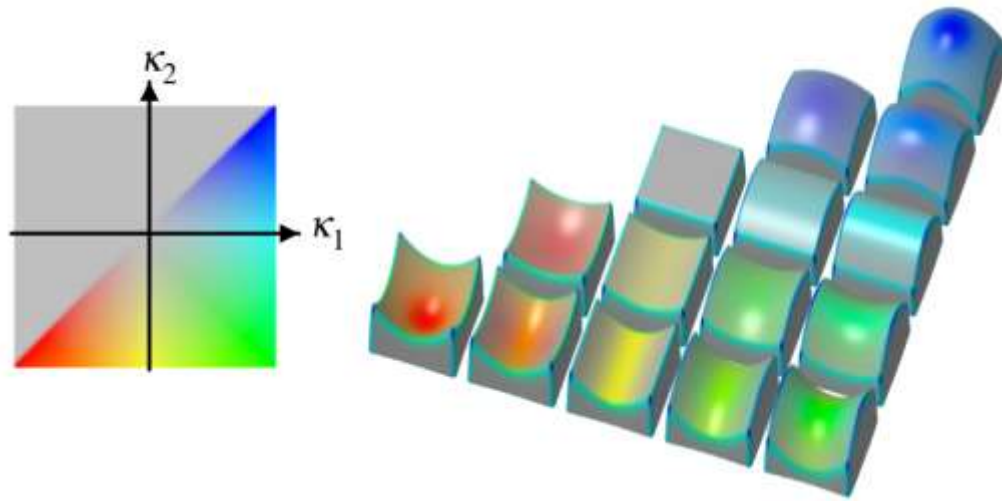
3D Hessians on 2D Slice



3D Hessians on Isosurface

Example: Geometry Tensors

- Encode principal curvature directions of a surface (convex vs. concave), as well as normal direction (zero eigenvalue)
 - Second-order, symmetric, indefinite



Coloring of Principal Curvatures

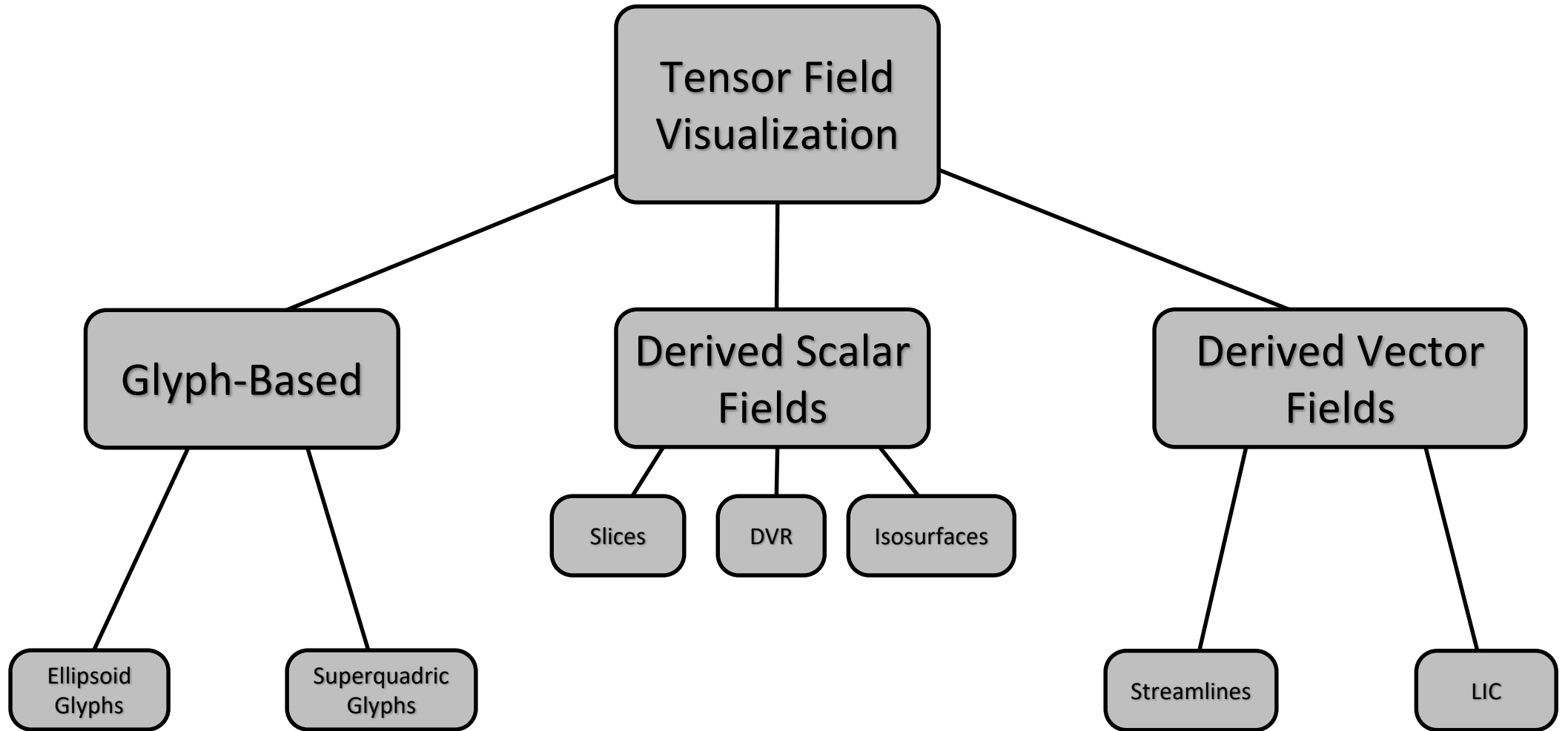


[Zhang et al. 2006]

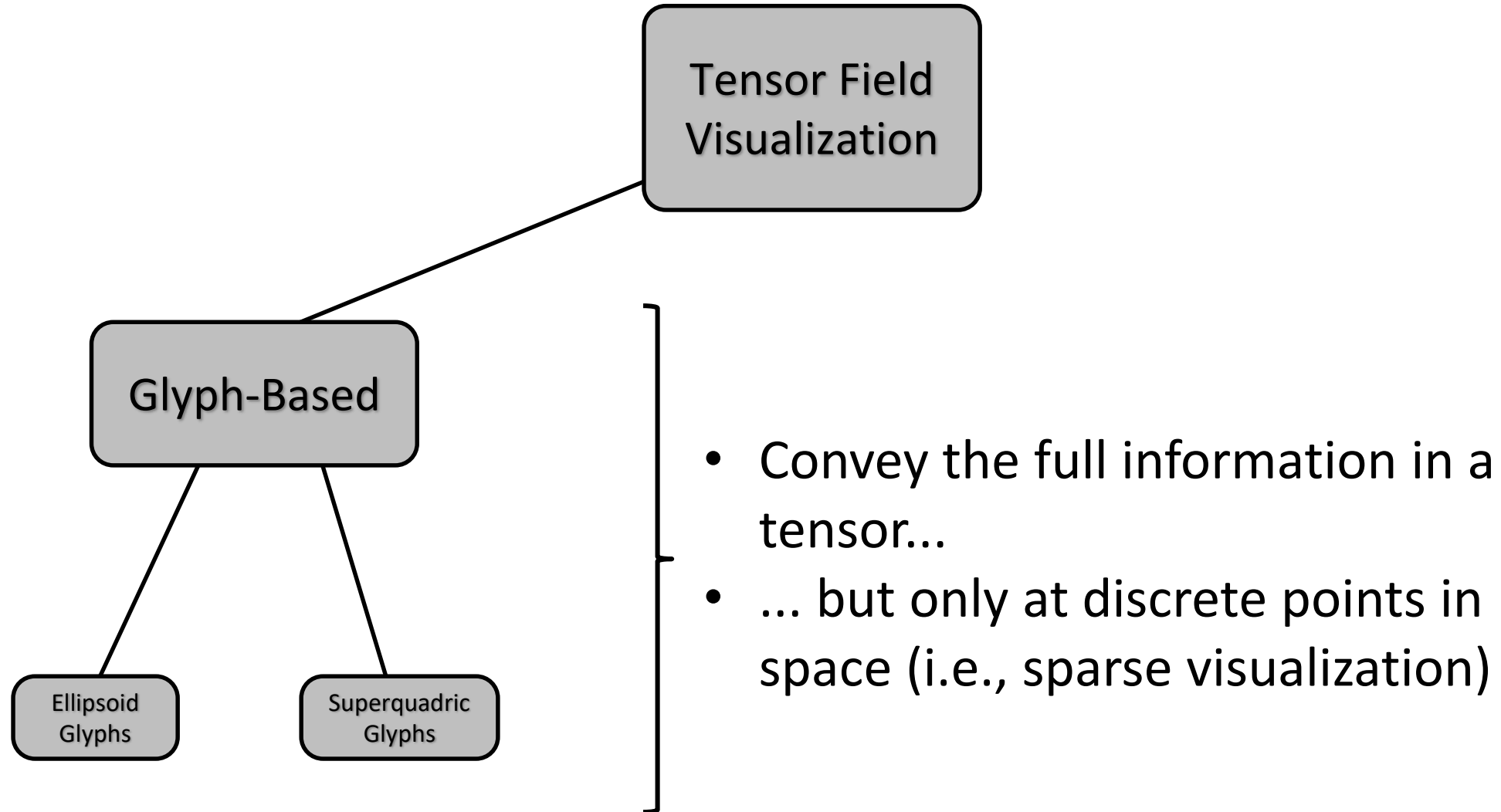
Summary: Introduction to Tensors

- **Tensors** (in our sense) are linear maps between vector spaces
 - Can be represented as matrices
 - Differ with respect to symmetry, definiteness
- **Applications** include
 - Diffusion Tensor Imaging (DTI)
 - material science (e.g., stress tensors, NLCs)
 - flow fields (Jacobians)
 - geometry processing (curvature)

Overview of Tensor Field Visualization

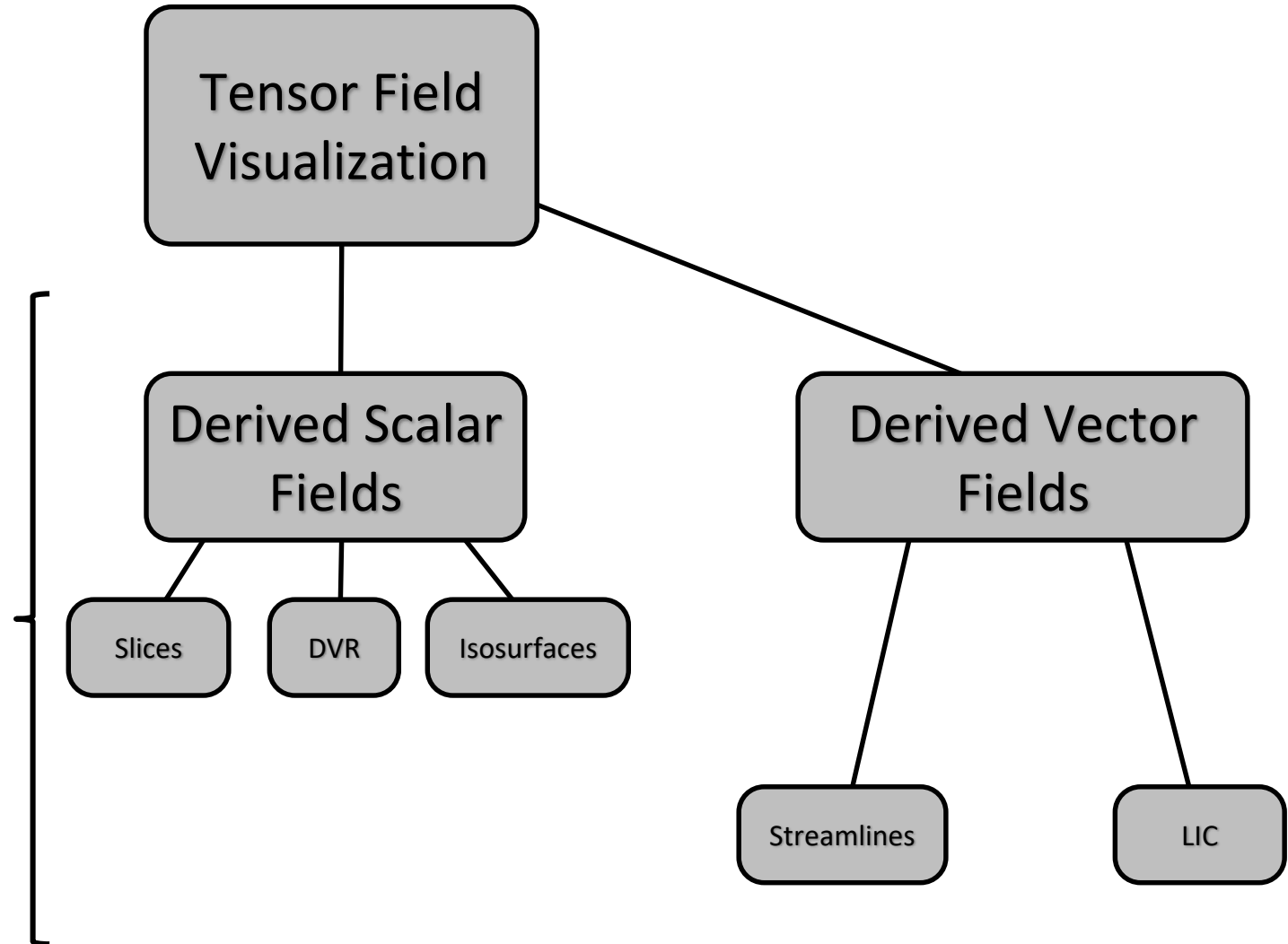


Overview of Tensor Field Visualization



Overview of Tensor Field Visualization

- Reduce information content
- Can be used to derive dense visualizations



Section 14.2: Tensor Glyphs

Reminder: Eigenvectors and Spectral Decomposition

- A nonzero vector $\mathbf{v}$ is an **eigenvector** of matrix $\mathbf{M}$ if the matrix performs a pure scaling by the corresponding **eigenvalue** λ :

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$$

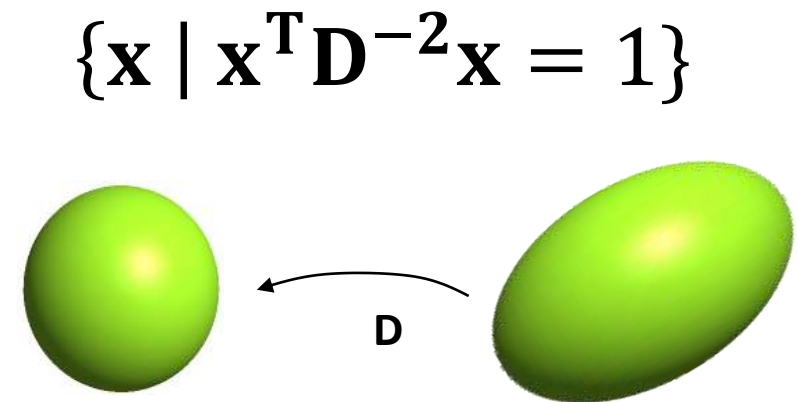
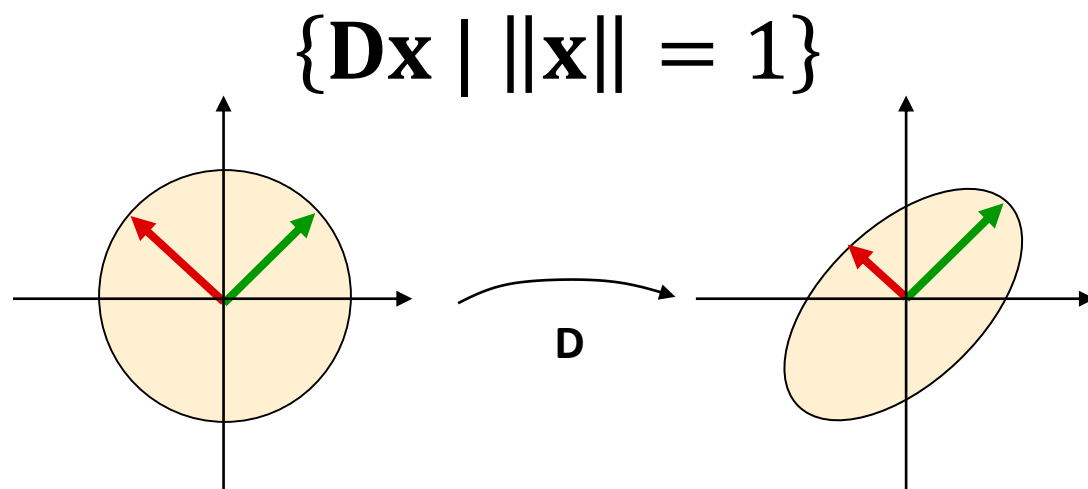
- The **spectral decomposition** of real symmetric matrices is in terms of real eigenvalues and orthogonal eigenvectors:

$$\mathbf{M} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$$

- Columns of $\mathbf{Q}$ are the eigenvectors
- Diagonal entries of $\mathbf{\Lambda}$ are the eigenvalues

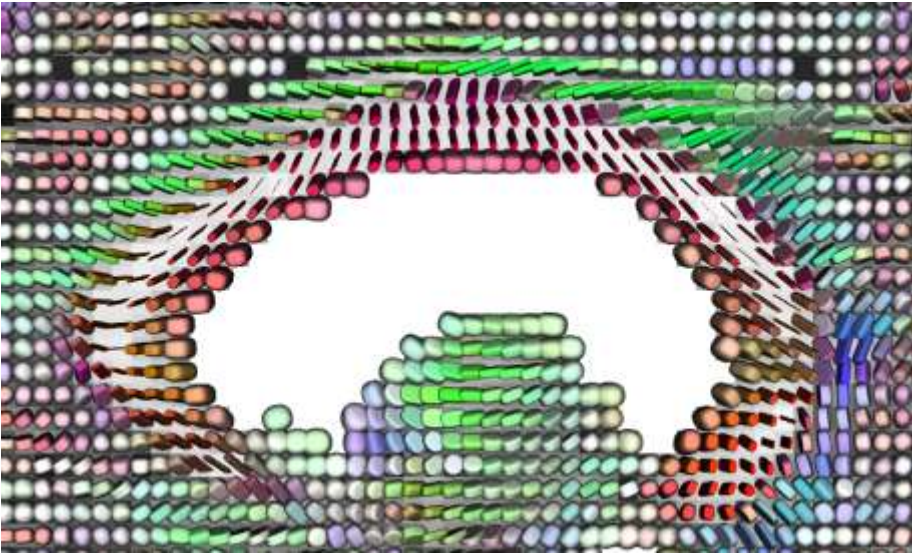
Tensor Ellipsoids

- Symmetric positive definite tensors $\mathbf{D}$ are naturally visualized by ellipsoids whose
 - axis directions are given by eigenvectors
 - half-axis lengths are given by the corresponding eigenvalues
- These ellipsoids can equivalently be written as

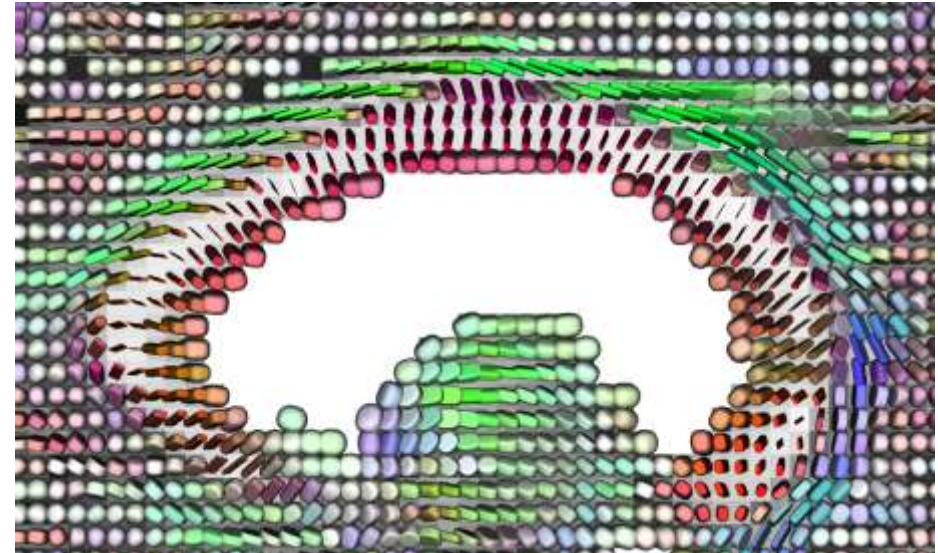


Glyphs in Diffusion MRI

- Glyphs are useful to
 - Quickly assess data quality
 - Identify artifacts
 - Inspect the data “immediately”, without complex and parameter-laden visualization algorithms



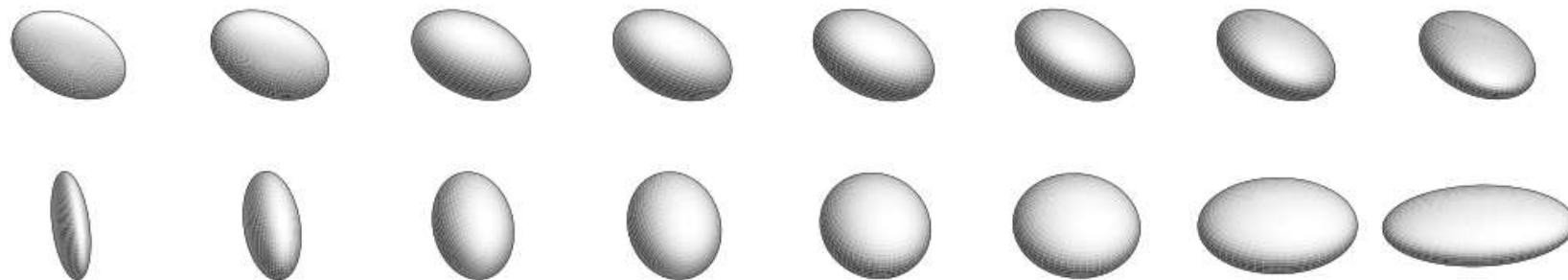
Flipped z Coordinate



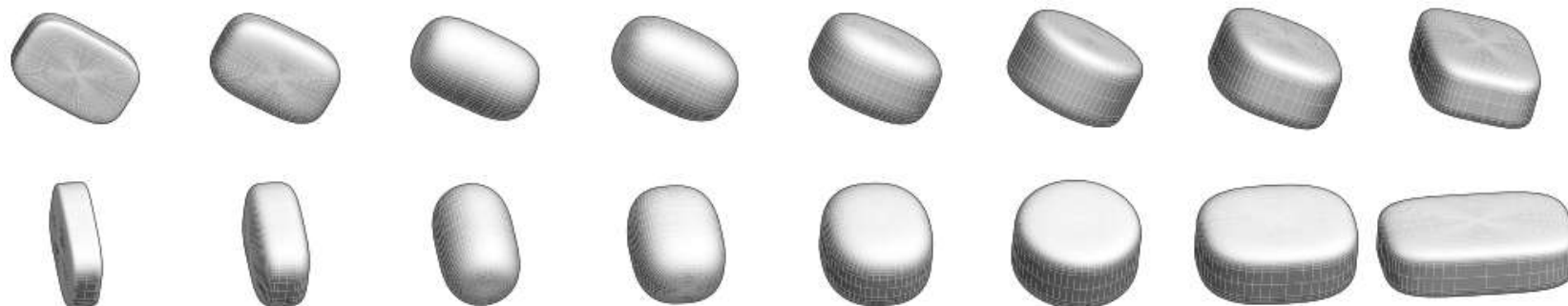
Corrected Sign

Superquadric Tensor Glyphs

- Ellipsoids suffer from **visual ambiguities**:



- Superquadric Glyphs** greatly reduce them:



Principles for Tensor Glyph Design

Faithful and expressive visualization requires:

- **Preservation of Symmetry:** Glyph should have the same symmetries as the tensor

$$\mathbf{D} = \mathbf{T} \mathbf{D} \mathbf{T}^{-1} \iff G(\mathbf{D}) = \mathbf{T} G(\mathbf{D})$$

- **Continuity:**

$$\mathbf{D}_1 \approx \mathbf{D}_2 \iff \text{appearance}(G(\mathbf{D}_1)) \approx \text{appearance}(G(\mathbf{D}_2))$$

- **Disambiguity:**

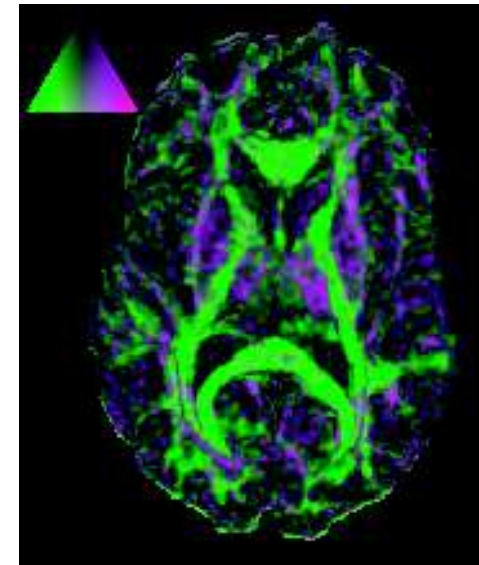
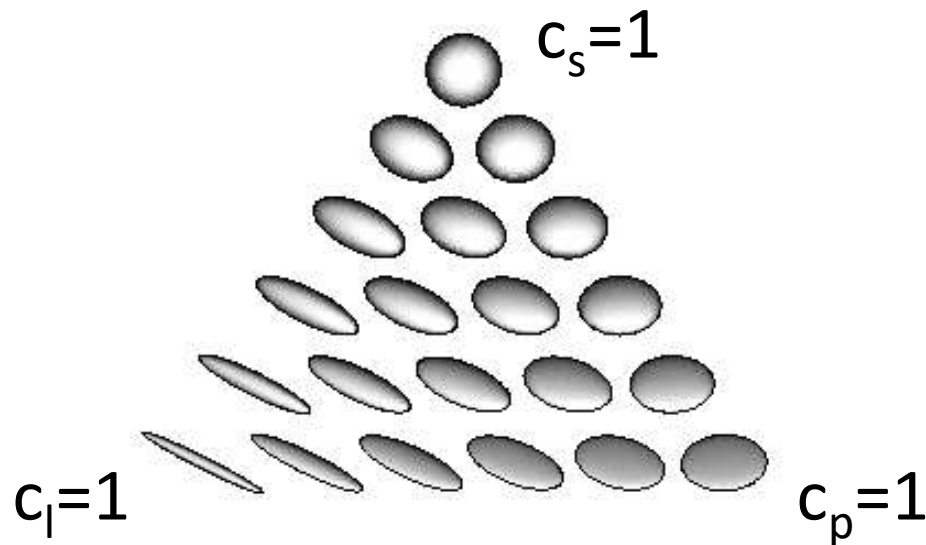
$$\mathbf{D}_1 \neq \mathbf{D}_2 \iff \text{appearance}(G(\mathbf{D}_1)) \neq \text{appearance}(G(\mathbf{D}_2))$$

Westin Measures

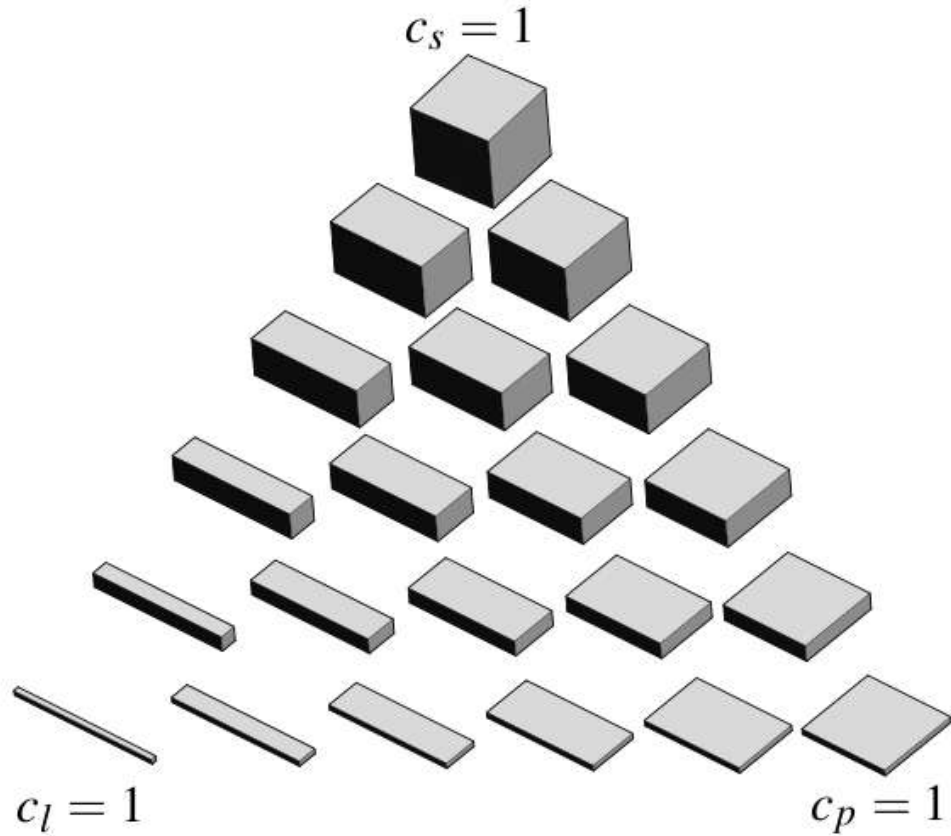
- Three rotationally invariant metrics based on sorted eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$
- Quantify amount of linearity / planarity / sphericity [Westin et al., 1997]

$$c_l = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} \quad c_p = \frac{2(\lambda_2 - \lambda_3)}{\lambda_1 + \lambda_2 + \lambda_3} \quad c_s = \frac{3\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3}$$

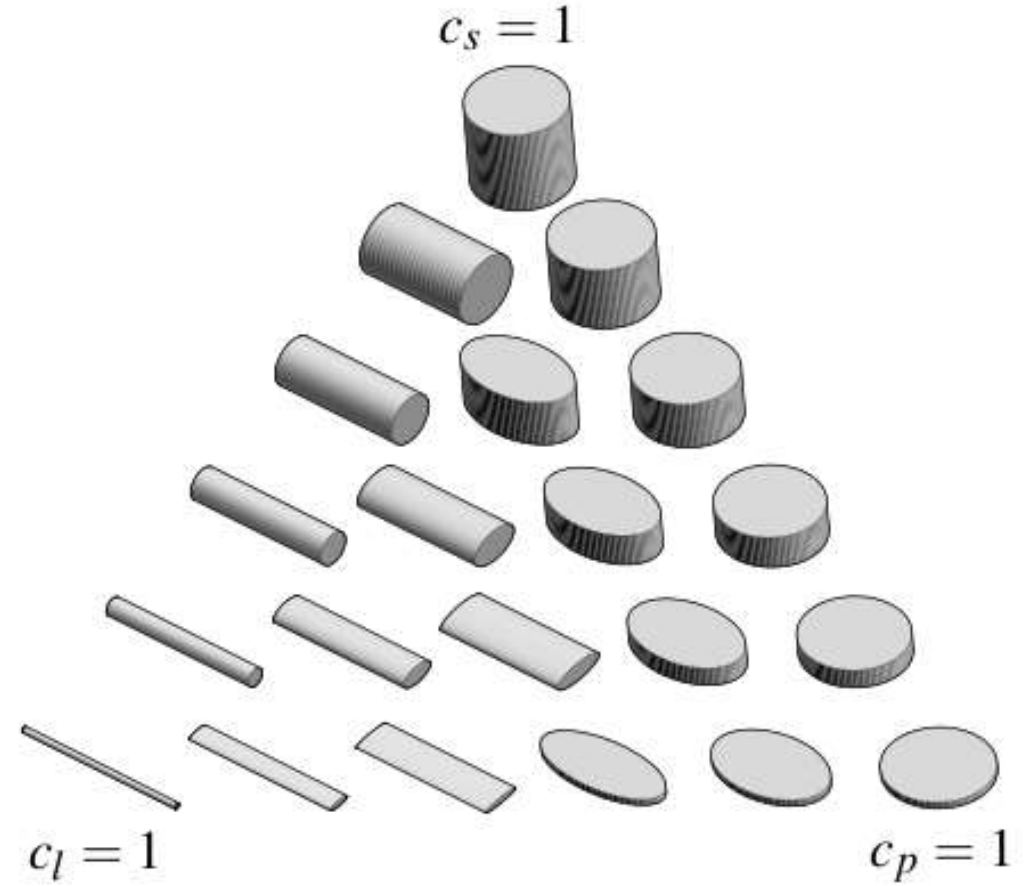
- Range in $[0,1]$, add up to one, thus define a barycentric space of tensor shapes



Limitations of Box and Cylinder Glyphs



Box Glyphs



Cylinder Glyphs

Superellipses in Architecture and Design



Fountain at Sergels Torg, Stockholm



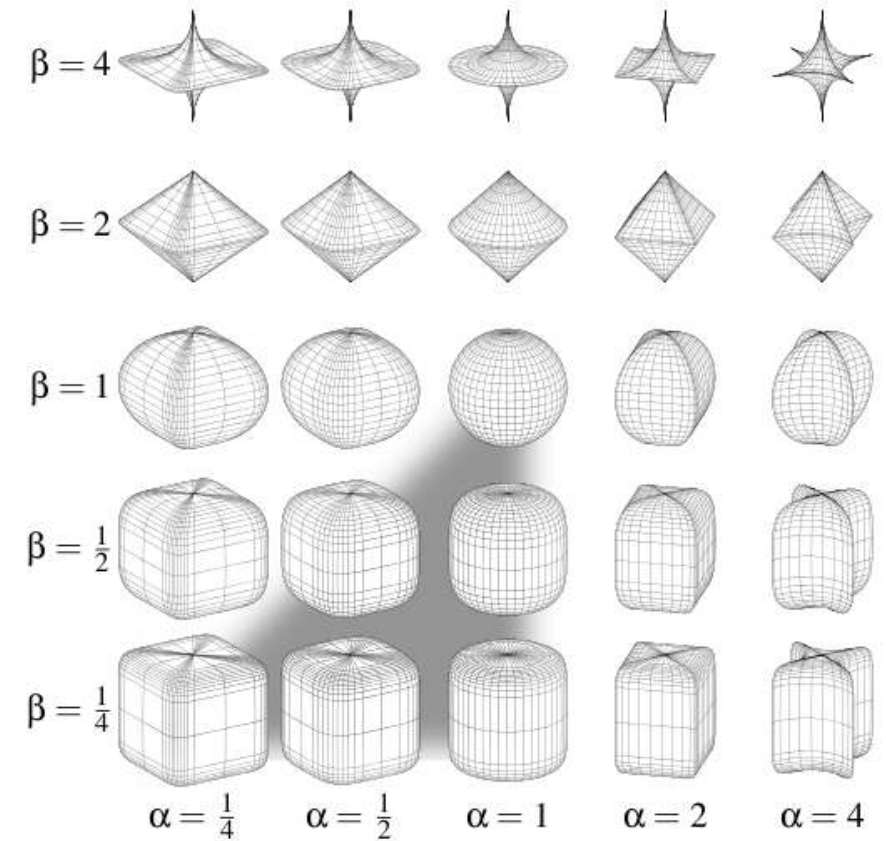
Superquadric Shape Space

- Superquadrics are given as

$$\mathbf{q}_z(\theta, \phi) = \begin{pmatrix} \cos^\alpha \theta \sin^\beta \phi \\ \sin^\alpha \theta \sin^\beta \phi \\ \cos^\beta \phi \end{pmatrix}, \quad \begin{matrix} 0 \leq \phi \leq \pi \\ 0 \leq \theta \leq 2\pi \end{matrix}$$

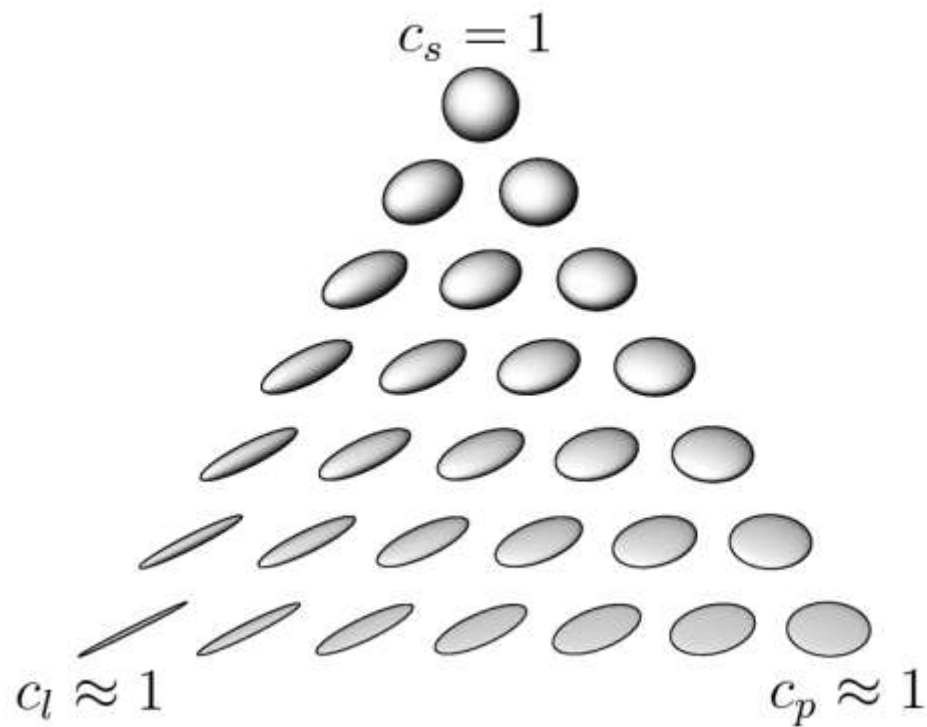
where exponentiation is defined to be sign preserving

$$\cos^\alpha \theta := \text{sign}(\cos(\theta)) \cdot |\cos \theta|^\alpha$$

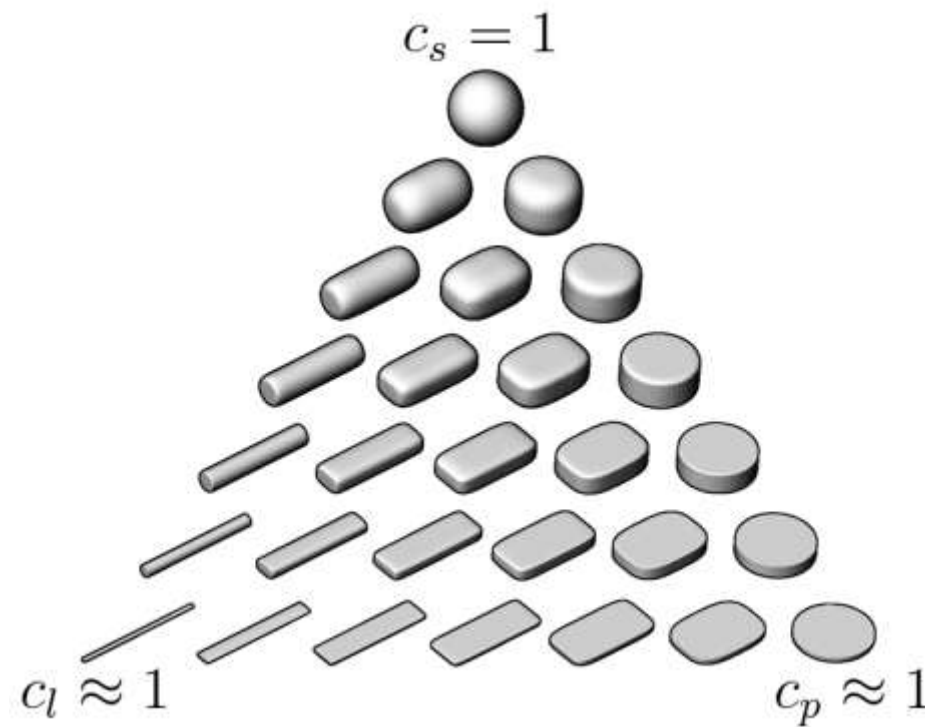


The Idea Behind Superquadric Glyphs

- **Ellipsoids** are transformations of the sphere
- **Superquadrics** smoothly interpolate between sphere, cylinder, and box



Ellipsoids



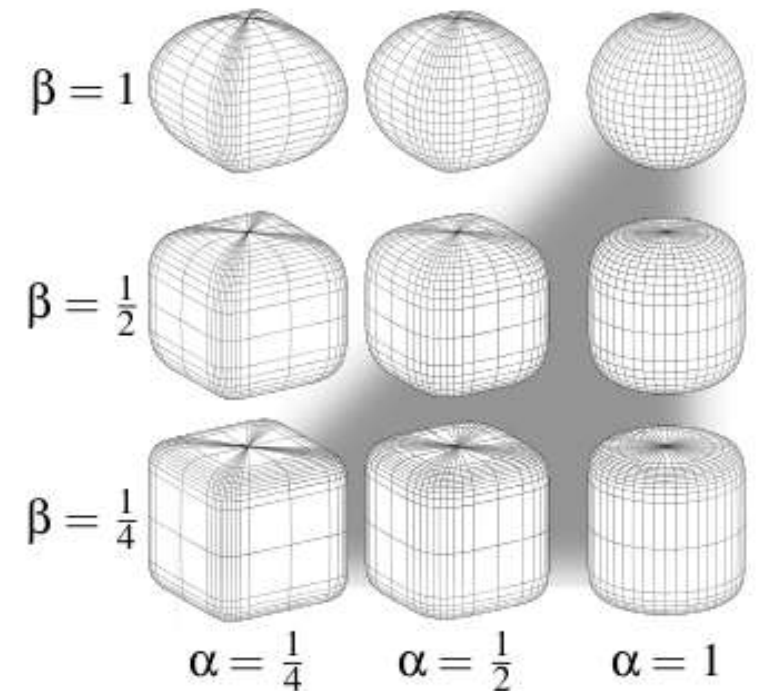
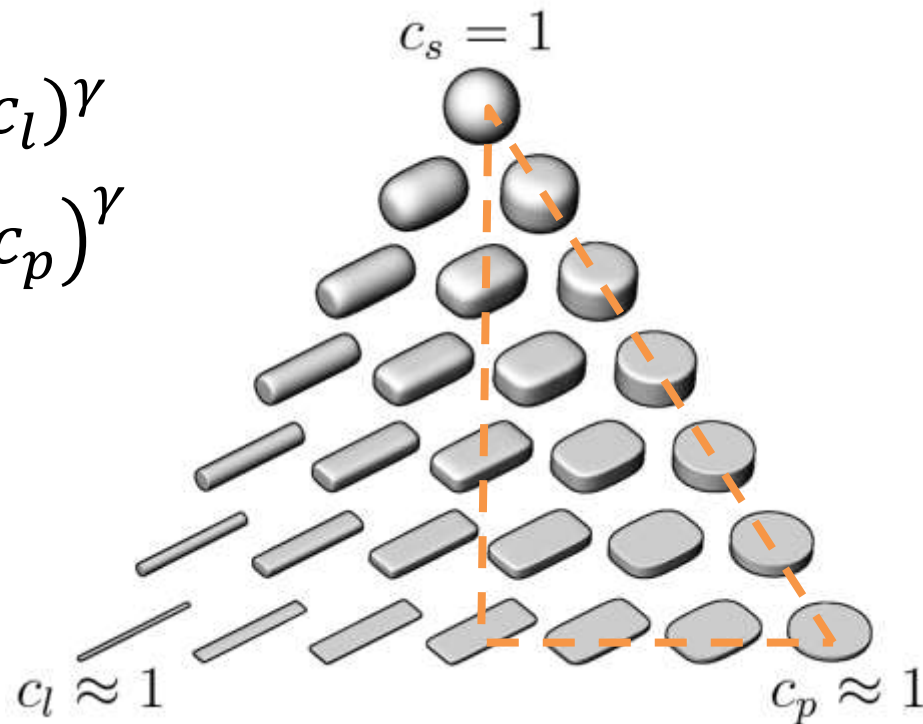
Superquadrics

Superquadric Parameters from Tensor Shape

- Assume x/y/z axes align with eigenvectors $\mathbf{e}_1/\mathbf{e}_2/\mathbf{e}_3$ for sorted eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$
- If $c_p > c_l$, we obtain the basic superquadric tensor glyph shape as follows:

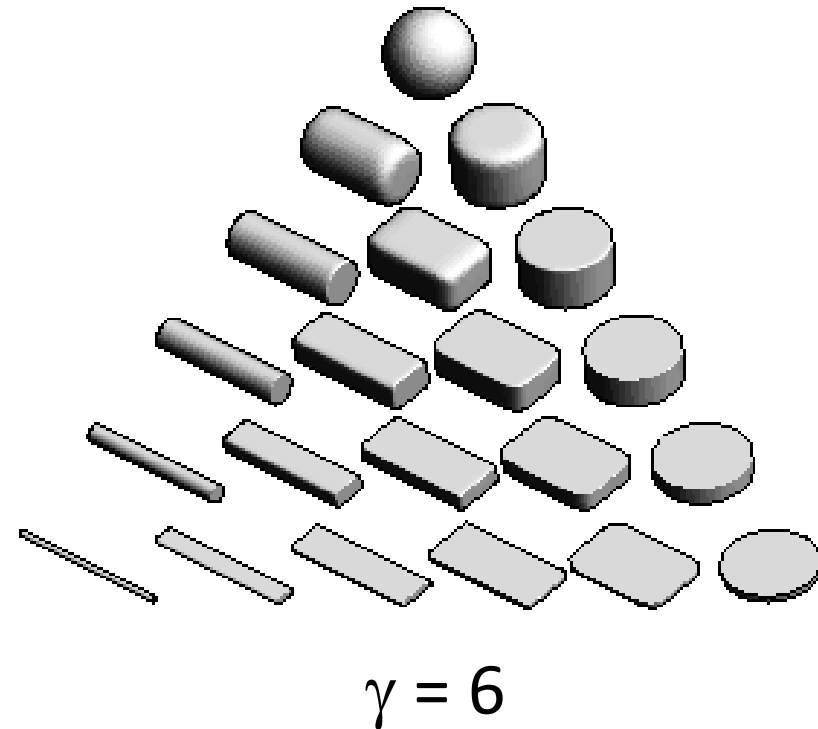
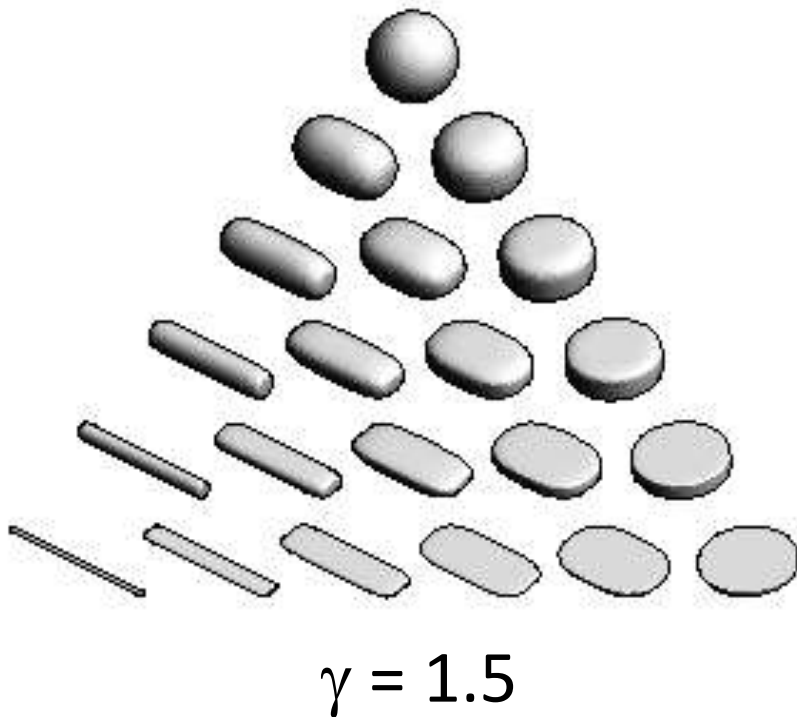
$$- \alpha = (1 - c_l)^\gamma$$

$$- \beta = (1 - c_p)^\gamma$$



Superquadric Glyphs: Sharpness

- Superquadric glyphs combine advantages of boxes, cylinders and ellipsoids
 - Edges give strong visual cues for orientation
 - Round shapes are used when orientation is not clearly defined
- “Sharpness” parameter γ decides how quickly sharp features develop



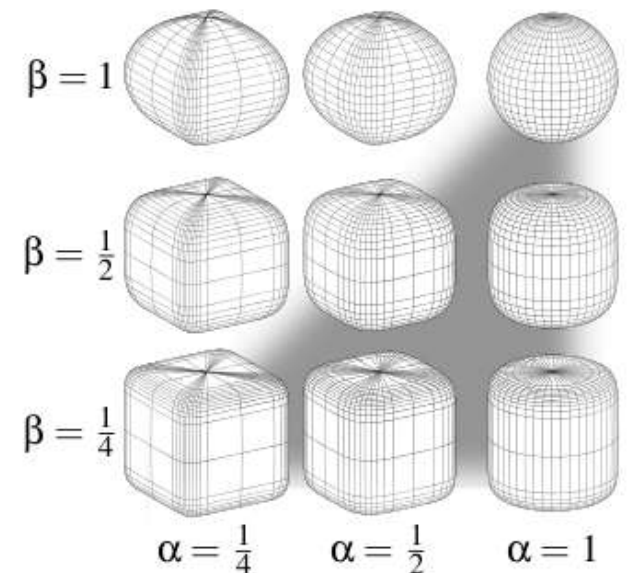
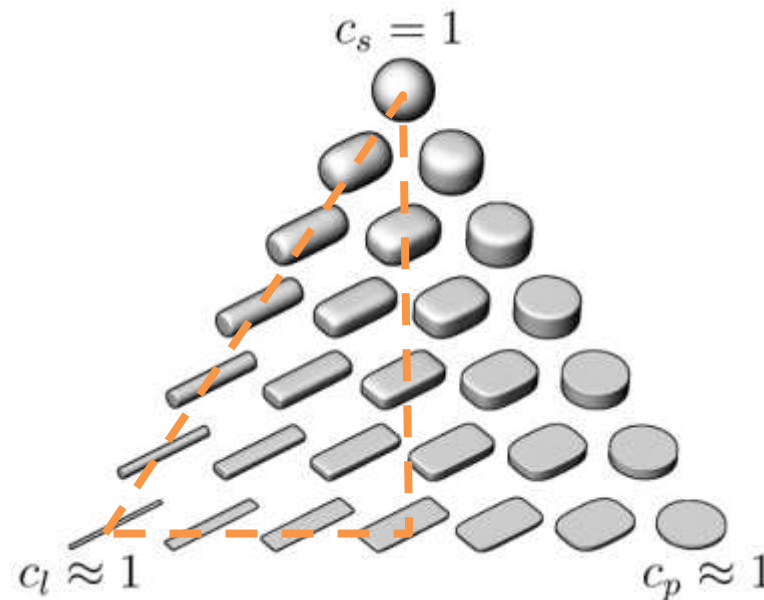
Superquadric Parameters when $c_l \geq c_p$

- If $c_l \geq c_p$, we use an alternative family of superquadrics that is defined to be symmetric around the x axis:

$$\mathbf{q}_x(\theta, \phi) = \begin{pmatrix} \cos^\beta \phi \\ -\sin^\alpha \theta \sin^\beta \phi \\ \cos^\alpha \theta \sin^\beta \phi \end{pmatrix}, \quad \begin{matrix} 0 \leq \phi \leq \pi \\ 0 \leq \theta \leq 2\pi \end{matrix}$$

$$- \alpha = (1 - c_p)^\gamma$$

$$- \beta = (1 - c_l)^\gamma$$



Seam in Parametrization

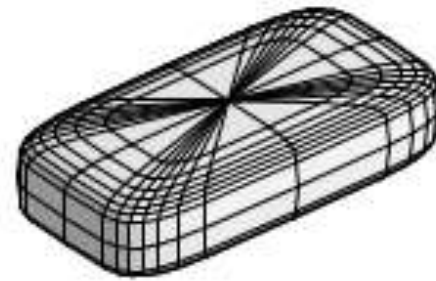
- Piecewise definition (depending on $c_l < c_p$ vs. $c_l \geq c_p$) creates **seam in parametrization**, but maintains continuity in surface geometry



$$c_l = 0.33$$
$$c_p = 0.27$$



$$c_l = 0.31$$
$$c_p = 0.29$$



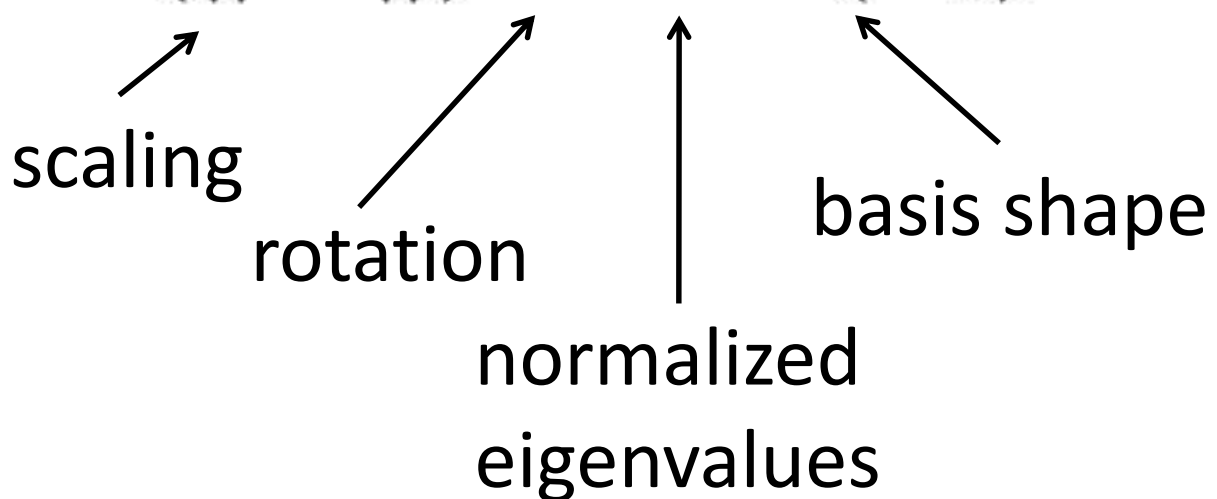
$$c_l = 0.29$$
$$c_p = 0.31$$



$$c_l = 0.27$$
$$c_p = 0.33$$

Superquadric Glyphs: Final Equation

To obtain final glyph, superquadric geometry is scaled by eigenvalues and rotated so that principal axes align with eigenvectors:

$$G(\mathbf{D}) = s(\|\mathbf{D}\|) \mathbf{R} \tilde{\Lambda} B(\tilde{\lambda}_i)$$


scaling

rotation

normalized eigenvalues

basis shape

Glyphs on Regular Grid

- Compared to glyphs on a regular grid...



Glyph Packing

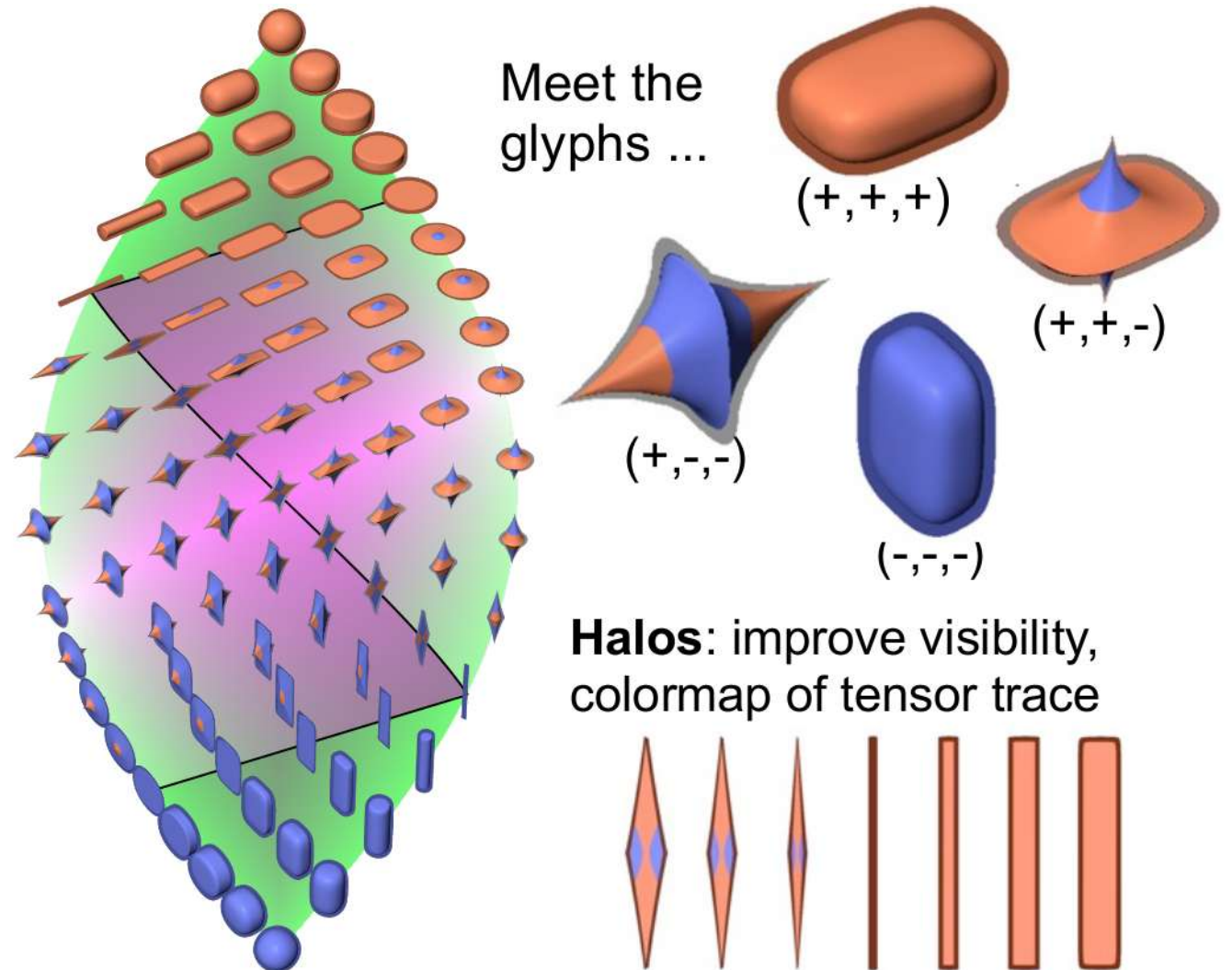
- **Packing glyphs** densely via a force-based layout and tensor field interpolation reduces occlusion and enhances perception of continuous structures



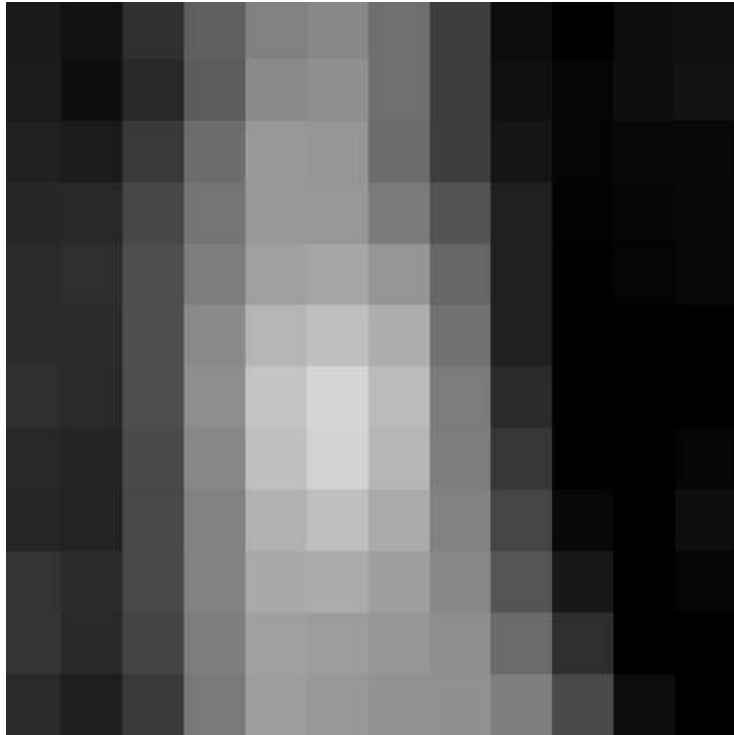
Outlook: Generalization to Indefinite Tensors

[Schultz/Kindlmann 2010]:

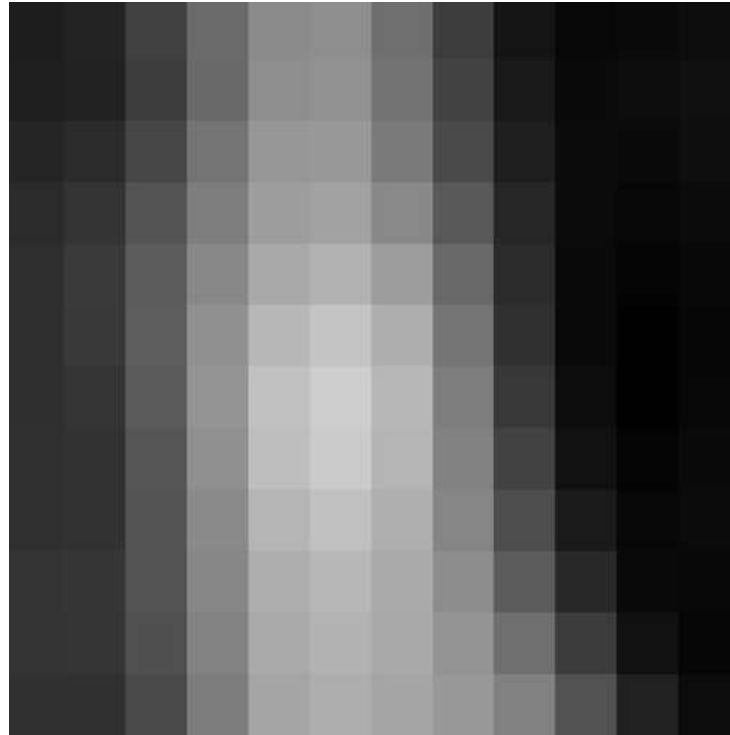
- *Color* indicates sign
- *Concavity* indicates differences in signs
 - Relies on a larger part of superquadric space
- *Halos* preserve visibility even when eigenvalues are close to zero



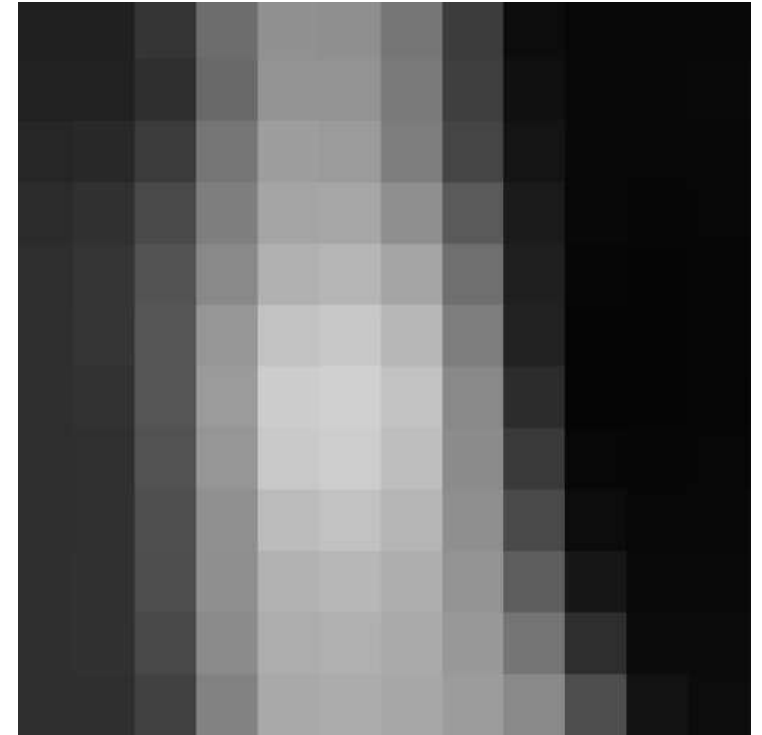
Inspecting Smoothing Image Filters



Original Data



Gaussian Smoothing

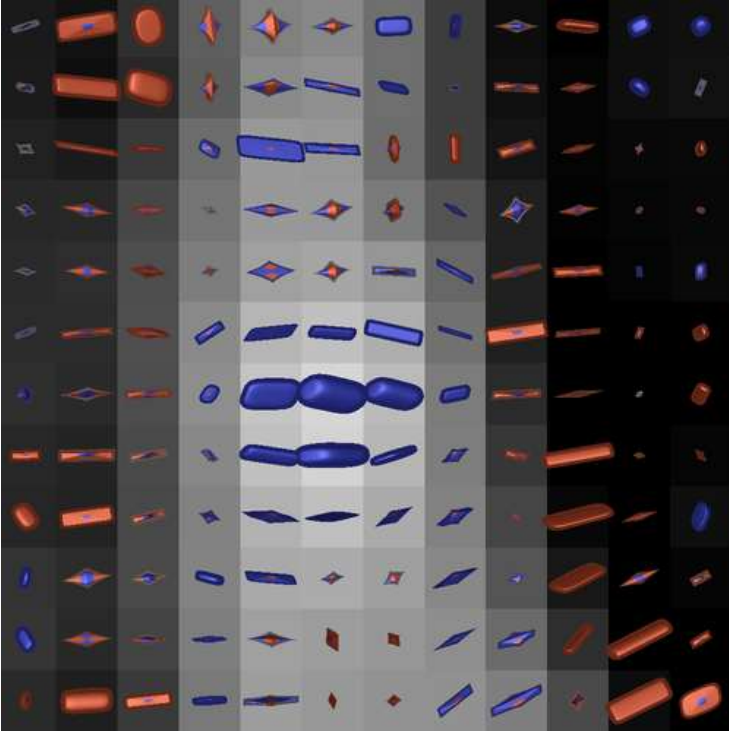


Total Variation

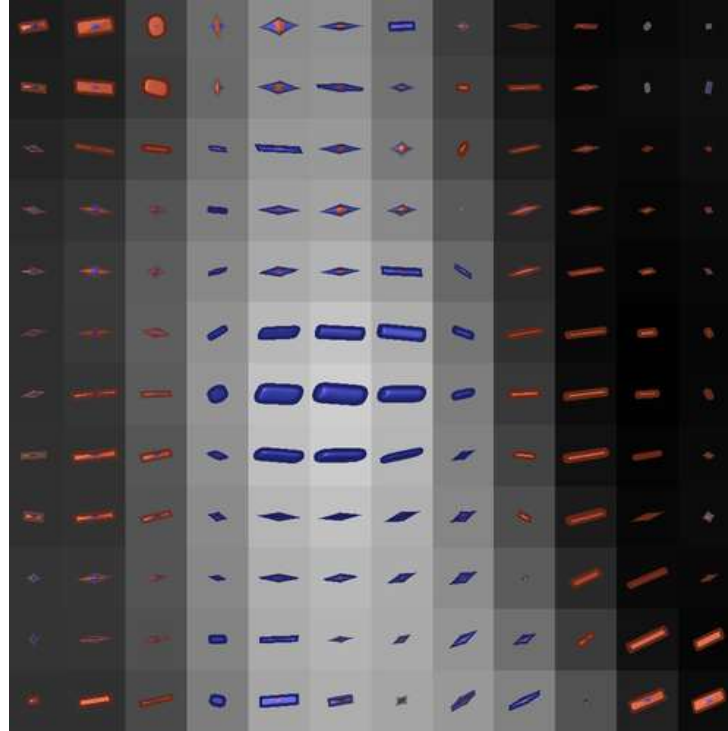
Equal Remaining Variance

What are Differences?

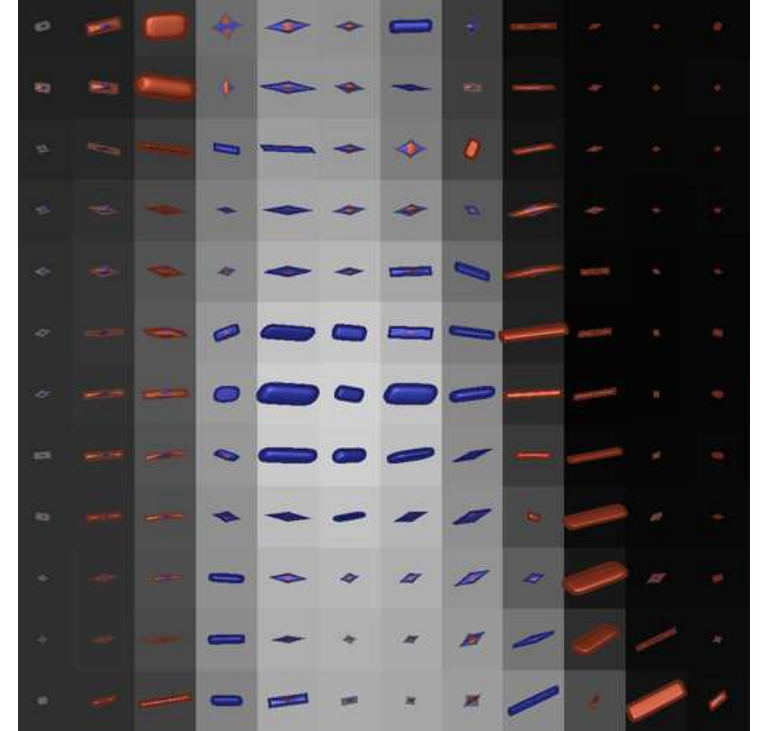
Inspecting Smoothing Image Filters



Original Data



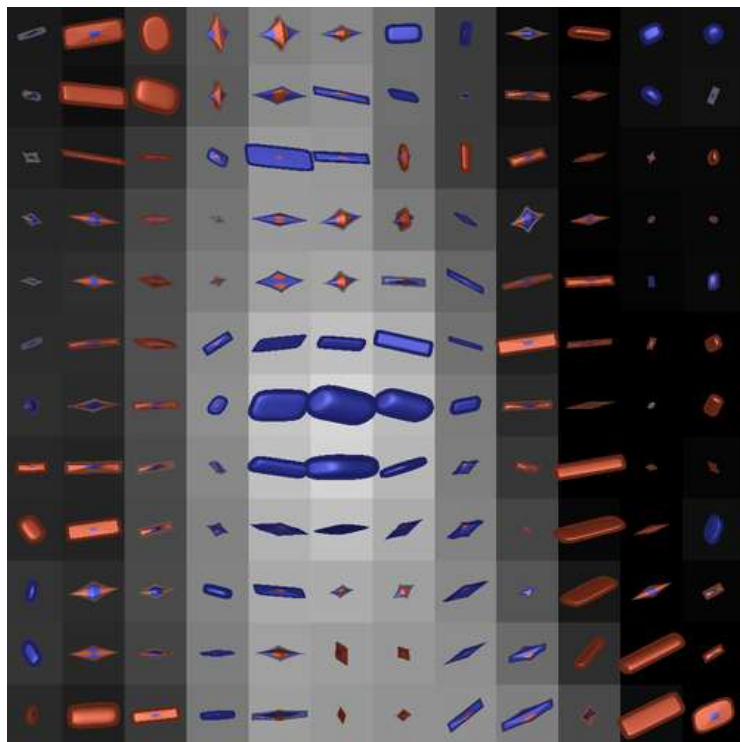
Gaussian Smoothing



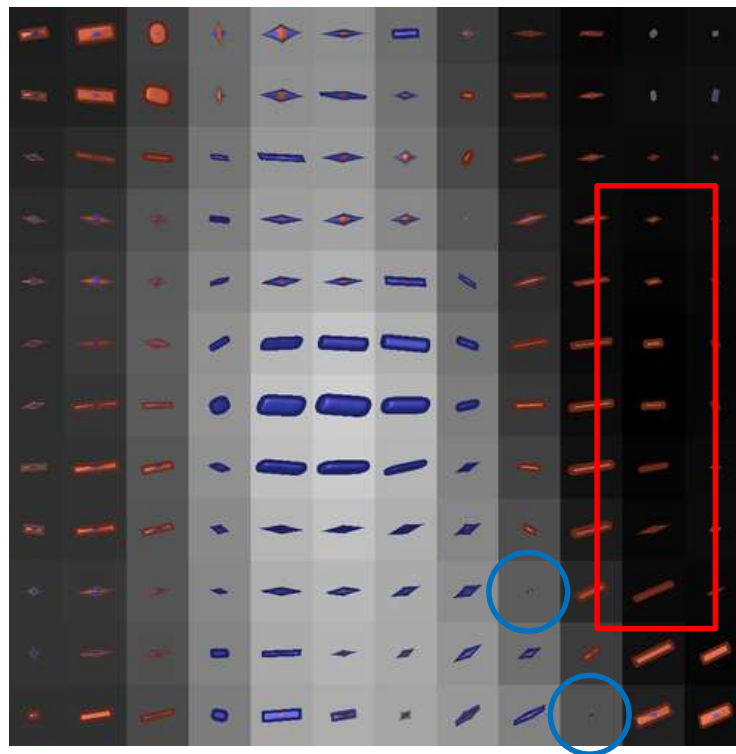
Total Variation Flow

Let's look at Hessians...

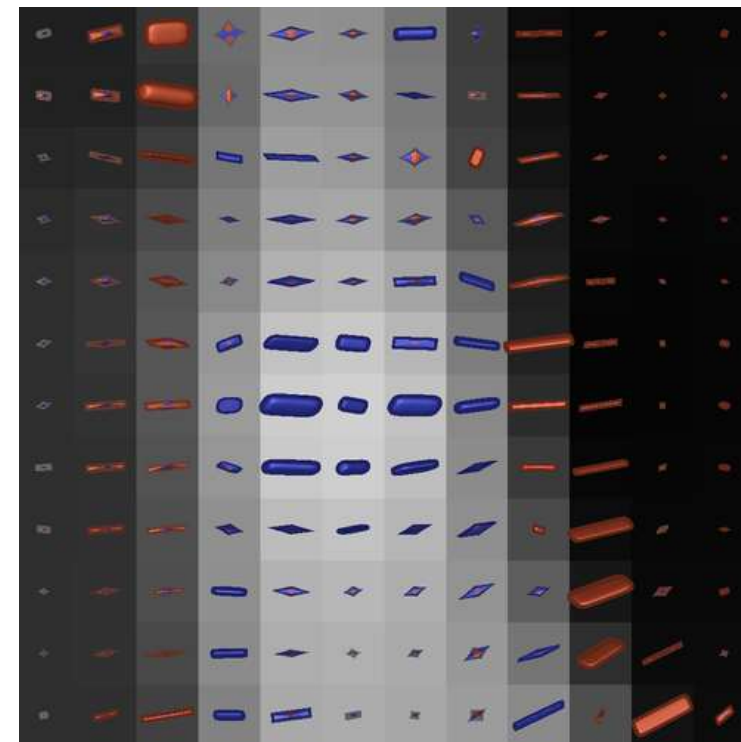
Inspecting Smoothing Image Filters



Original Data



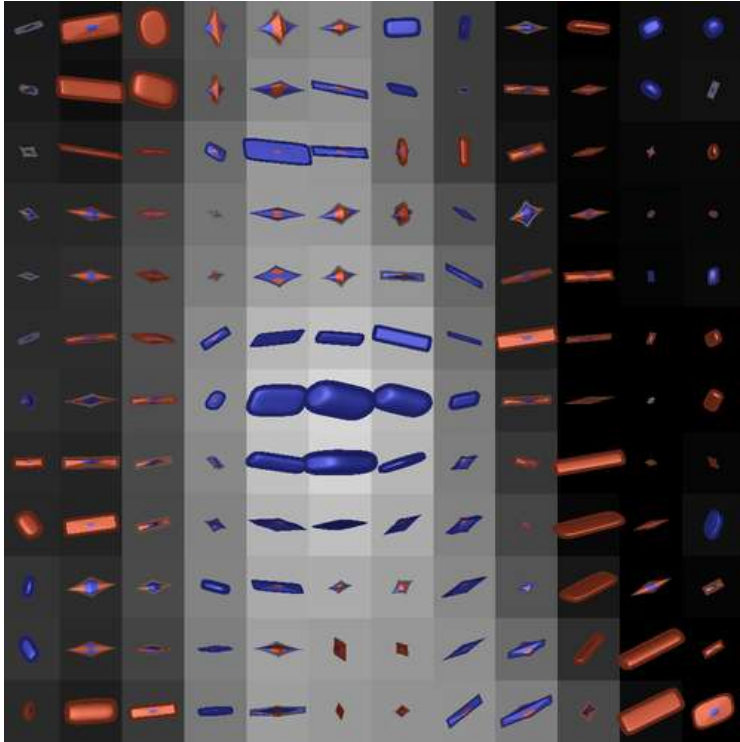
Gaussian Smoothing



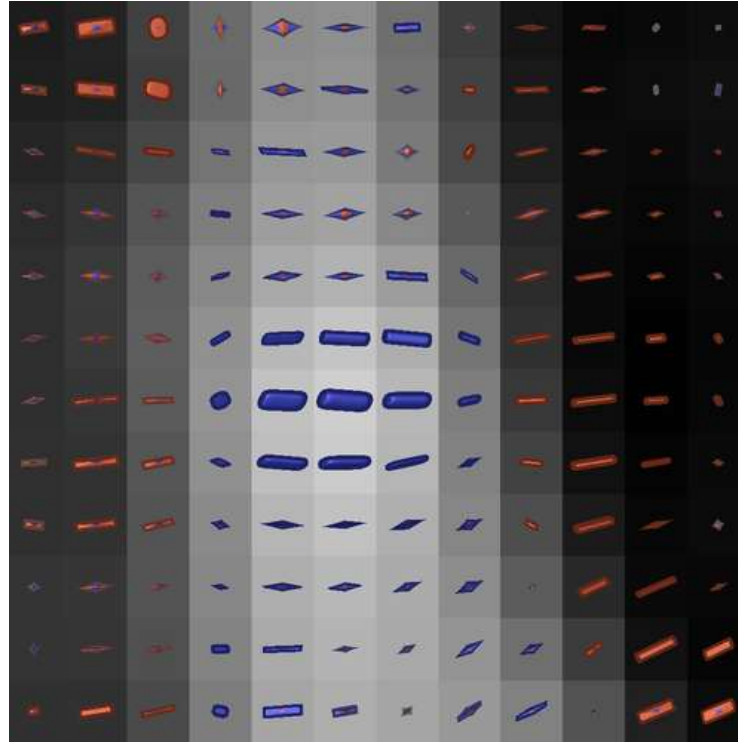
Total Variation Flow

Gaussian Filtering **Smooths Hessians.**
Leads to “**smearing out**” into flat regions
and **cancellation effects**.

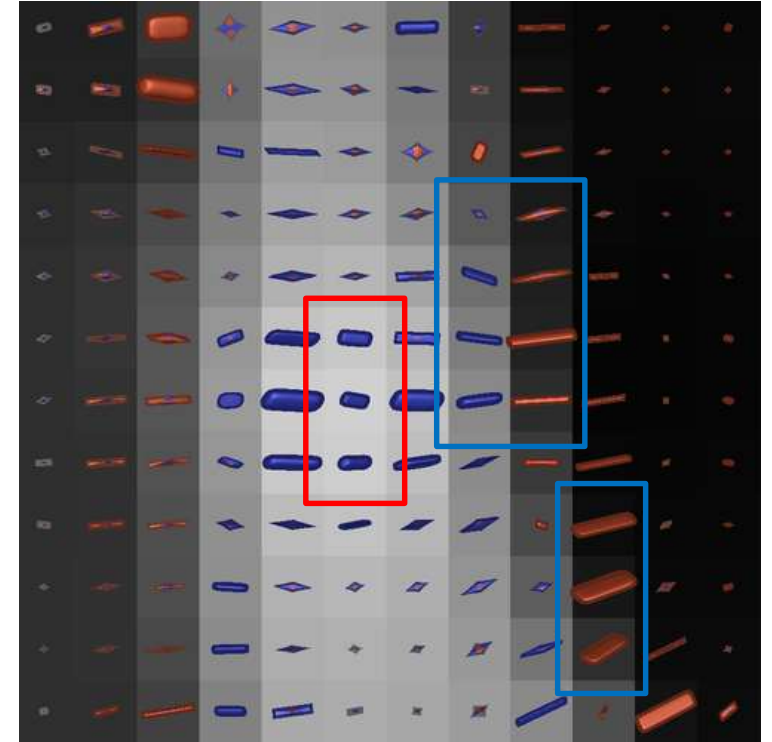
Inspecting Smoothing Image Filters



Original Data



Gaussian Smoothing

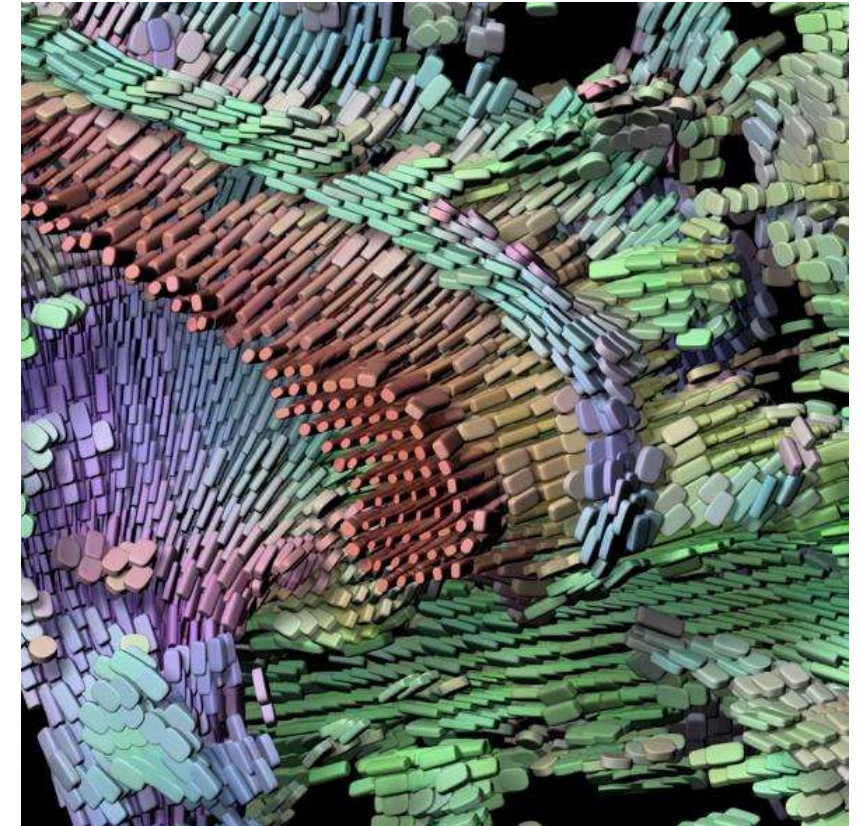


Total Variation Flow

TV flow creates **flat regions**
and **sharp edges** between them.

Glyphs: Benefits and Drawbacks

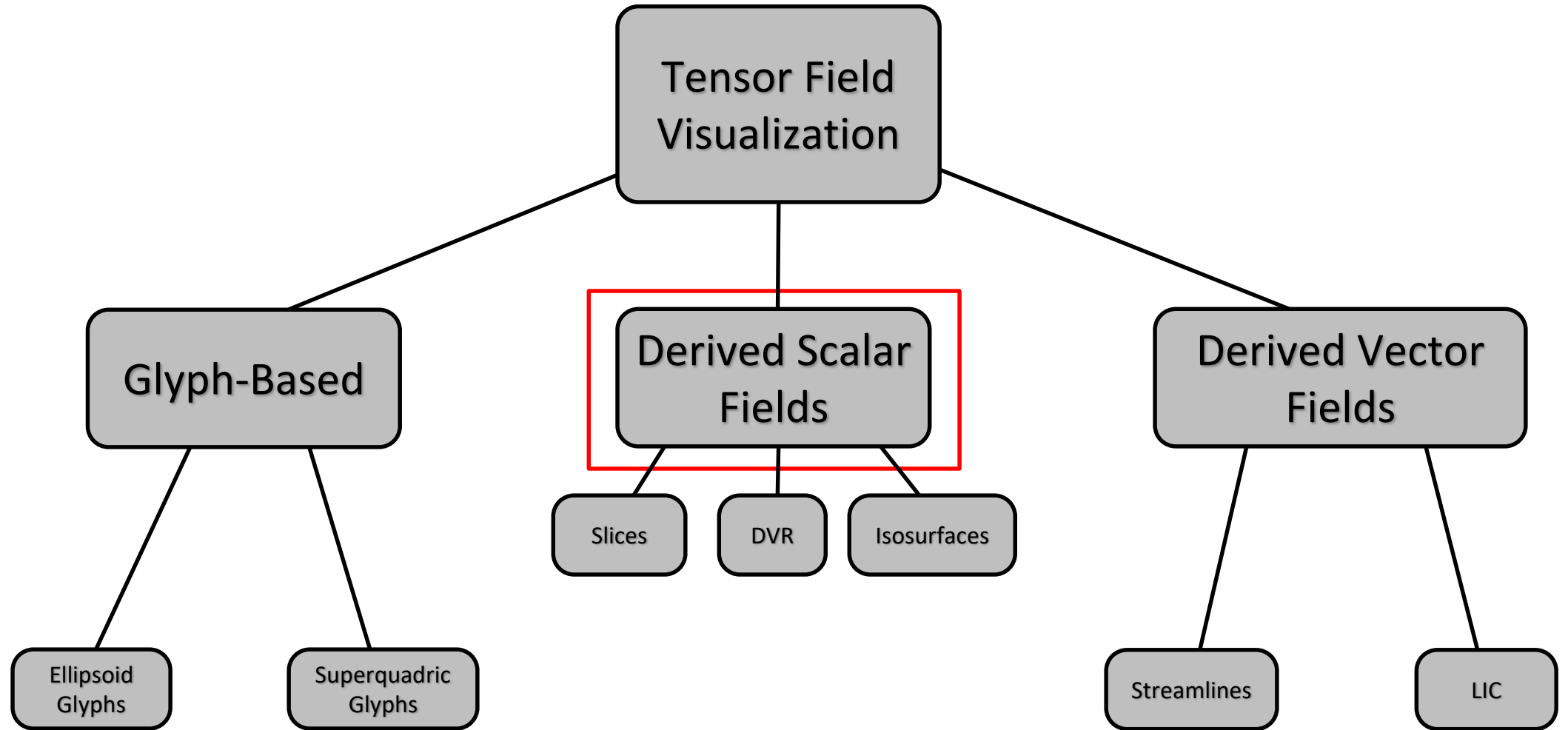
- Unlike scalar- or vector-based visualizations,
 - glyphs convey the full information present in the tensor
 - glyphs only show the field at discrete points in space
- Glyphs work well for 2D slices, but suffer from occlusion in 3D
- In 3D, culling of tensors is necessary, e.g. based on an anisotropy threshold
 - in the example, only show $c_l + c_p \geq 0.5$



Section 14.3:

Visualizing Derived Scalar Fields

Overview of Tensor Field Visualization



Scalar Invariants

- **Invariant** = scalar quantity that
 - is a function of the tensor
 - does not change under rotation
- **Eigenvalues** are invariant
 - All measures that can be expressed as functions of eigenvalues are invariant
 - Eigenvalues parameterize the “shape” of symmetric tensors
 - We assume sorting $\lambda_1 \geq \lambda_2 \geq \lambda_3$
- **Definitions and interpretation** are highly application specific
 - We will see examples from diffusion MRI and from stress tensors

Matrix Trace / Mean Diffusivity

- **Matrix Trace**

- Sum of diagonal elements:

- $$\text{tr}(\mathbf{T}) = T_{xx} + T_{yy} + T_{zz}$$

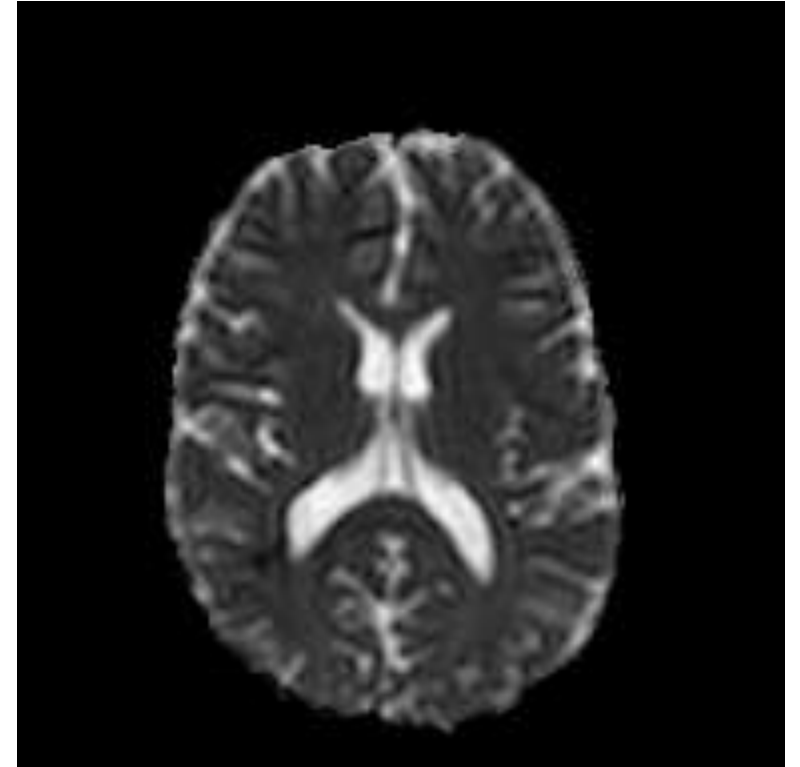
- For symmetric $\mathbf{T}$, same as sum of eigenvalues: $\lambda_1 + \lambda_2 + \lambda_3$

- **Mean Diffusivity (MD)**

- Average eigenvalue: $\text{tr}(\mathbf{T})/3$

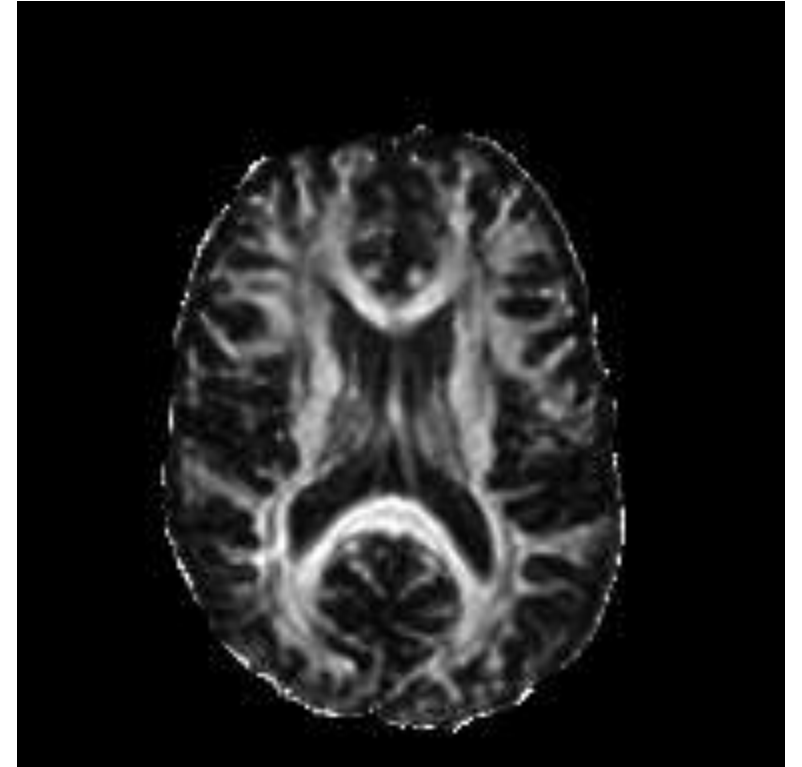
- Interpretation in DT-MRI:

- Average diffusivity (over all directions)
 - Approx. constant in healthy tissue, elevations indicate loss of tissue integrity
 - MD much higher in ventricles than in tissue



Fractional Anisotropy (FA)

- **Fractional Anisotropy**
 - Quantifies the degree of anisotropy
 - Interpretation in DT-MRI:
 - High values in white matter
 - Correlates with fiber density / integrity
 - Also correlates with orientation dispersion
 - Based on variance of eigenvalues
 - Normalized to [0,1] (if positive definite)

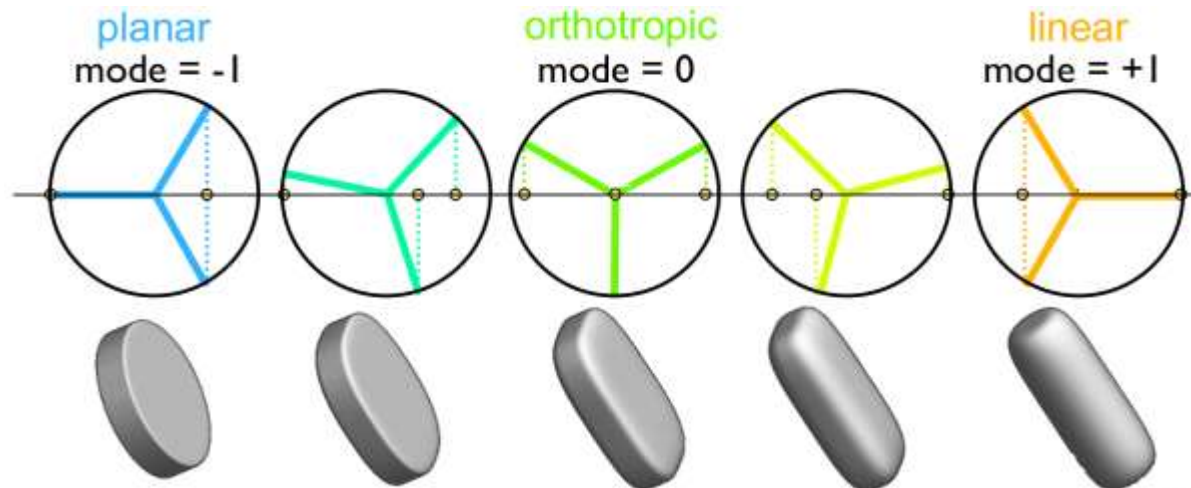
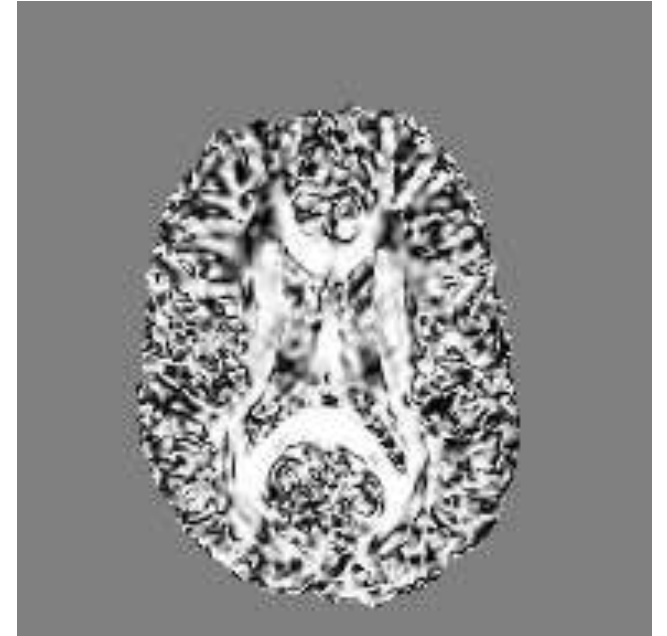


$$FA = \sqrt{\frac{3}{2}} \frac{\sqrt{(\lambda_1 - MD)^2 + (\lambda_2 - MD)^2 + (\lambda_3 - MD)^2}}{\sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}}$$

Tensor Mode

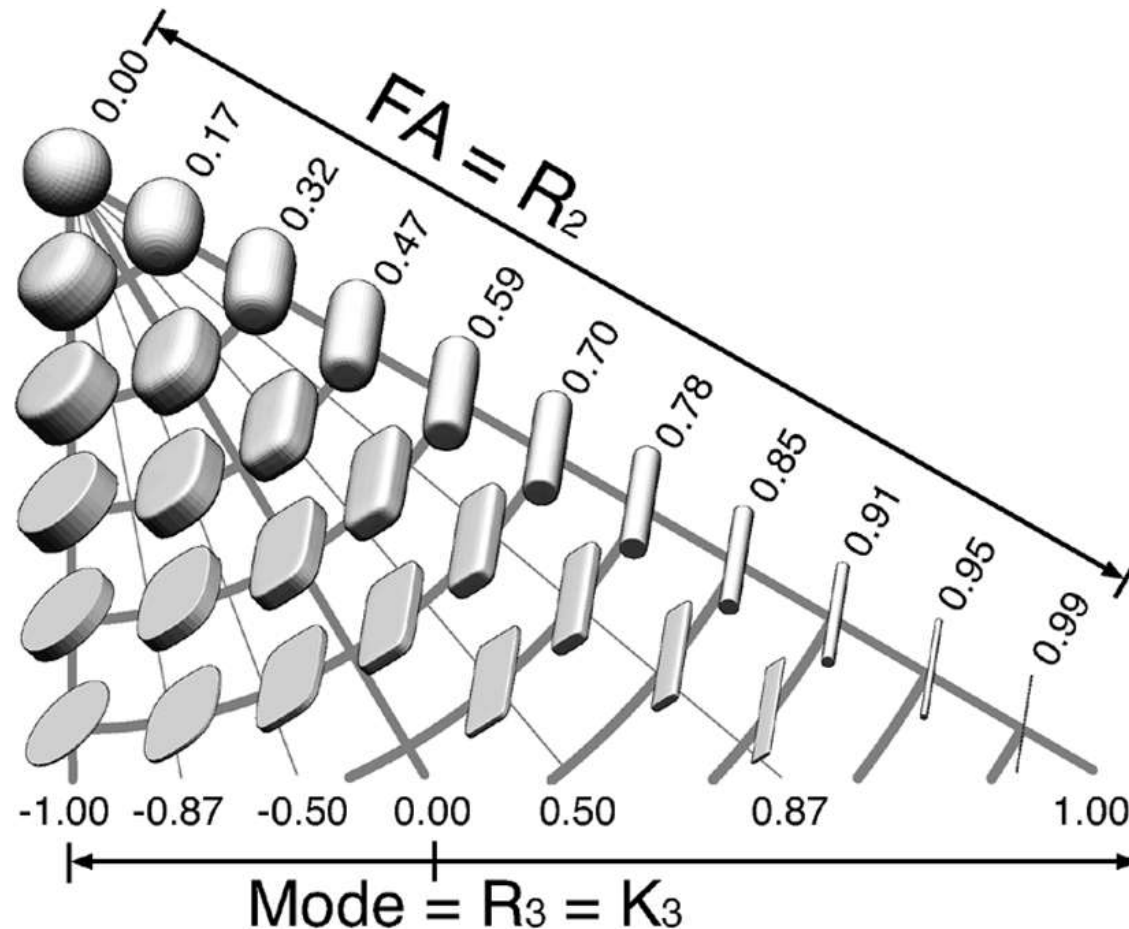
- **Tensor Mode**

- If $FA > 0$, tensor mode describes the *type* of anisotropy
- Planar: mode = -1 (black)
- Linear: mode = 1 (white)
- Orthotropic: mode = 0



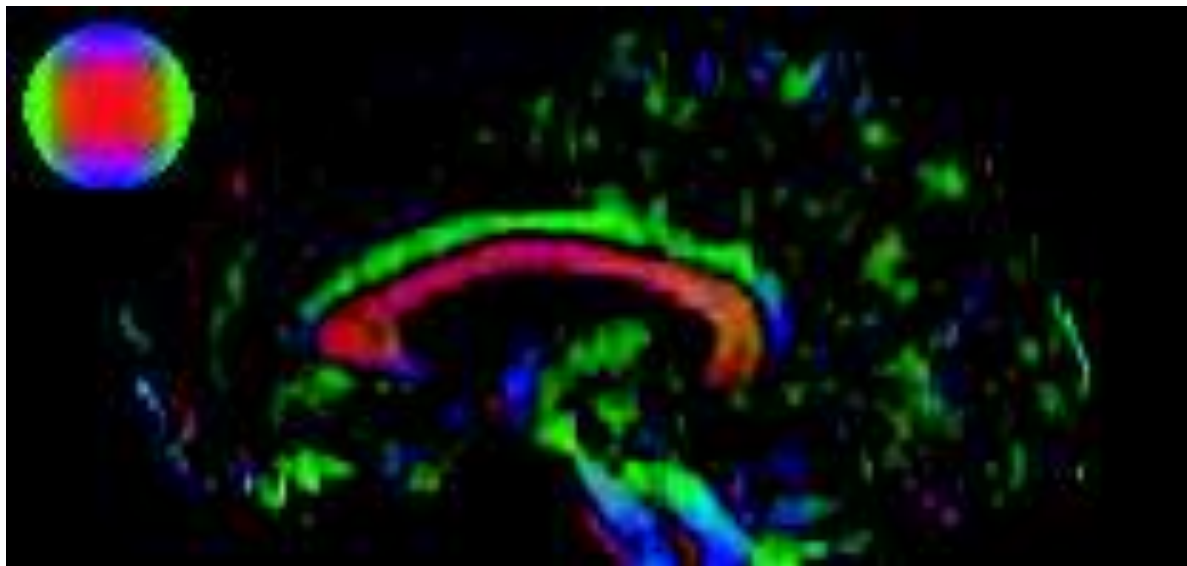
FA vs. Tensor Mode

- FA and mode are not fully independent
 - Maximum FA=1 only achieved with mode=1



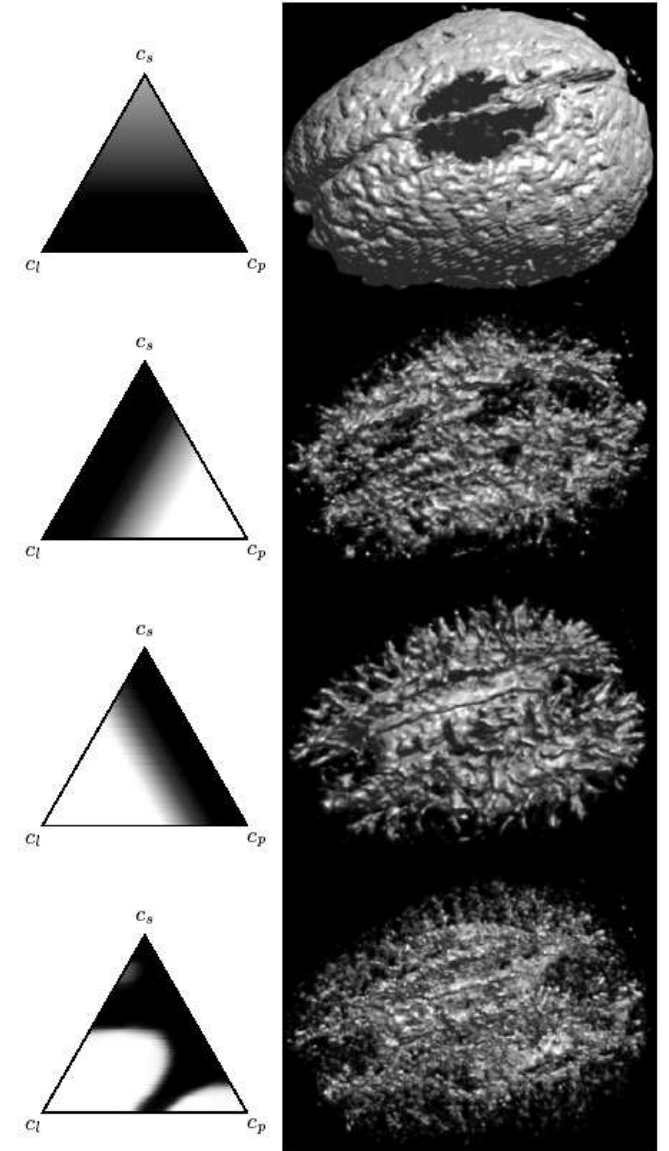
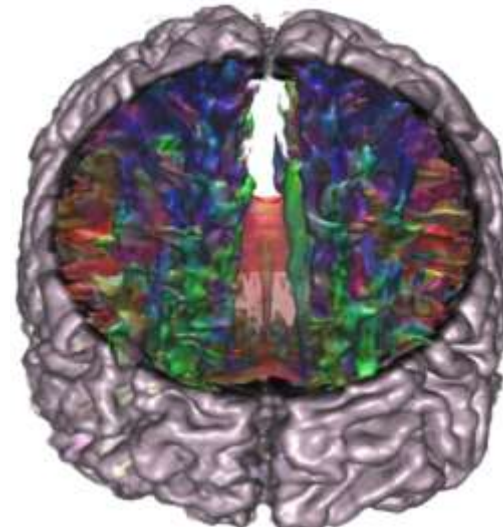
Color Coded Eigenvector Maps

- Standard Color Coding: XYZ->RGB
- Usually applied to principal eigenvector:
$$R = |v_{1x}|, \quad G = |v_{1y}|, \quad B = |v_{1z}|$$
- Usually modulated by FA or c_i
 - *Quiz:* Which of the two would you prefer and why?



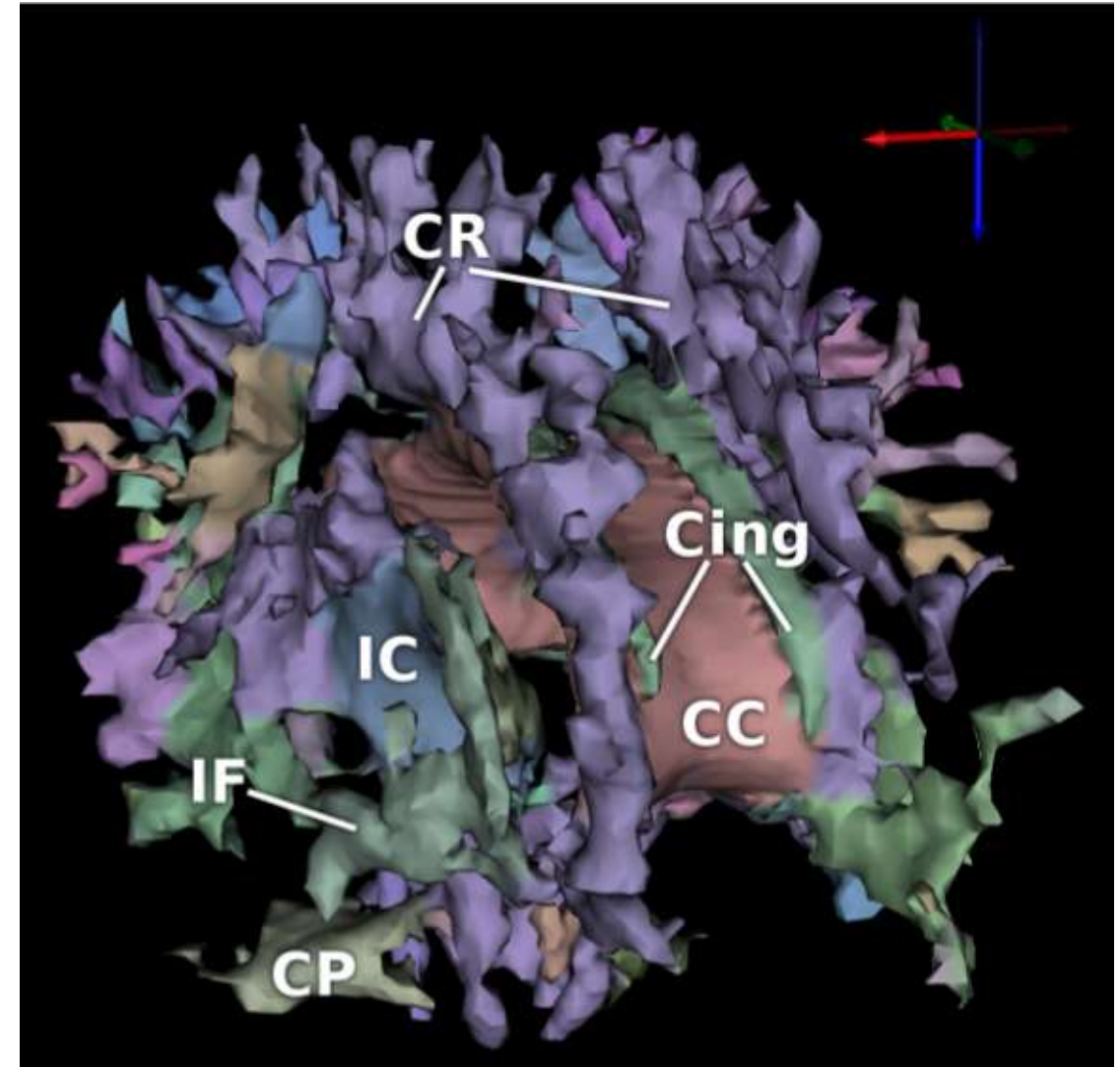
Basic Volume Rendering of Diffusion Tensor Fields

- Define opacity based on scalar measures such as Westin's c_l , c_p , c_s
- Color might be defined as on previous slide
- Shading based on gradient of opacity
 - Computation involves the chain rule
 - *Simple alternative*: Compute invariant(s) on a sufficiently fine grid and use standard DVR
- Combination with standard MRI can provide focus+ context (e.g., lens metaphor)



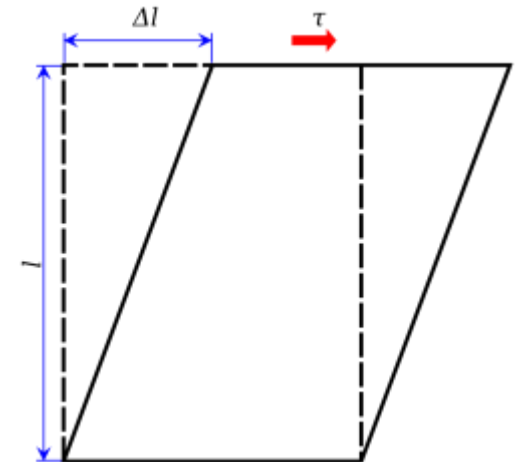
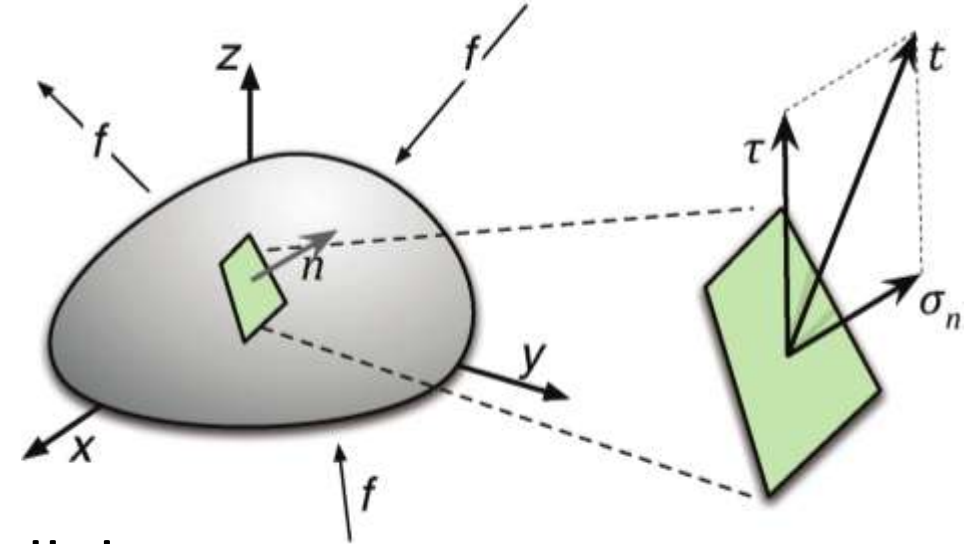
Isosurfaces in Tensor Fields

- *Example:* Isosurfaces of c_1 visualize the outline of the white matter core
 - Color indicates a segmentation into parts corresponding to different fiber bundles
- [Schultz et al. 2007]



Towards Stress Invariants: Normal and Shear Stress

- A stress tensor $\mathbf{T}$ maps a unit normal vector $\mathbf{n}$ to a traction vector $\mathbf{t}$ that describes the internal forces acting on the plane that is perpendicular to $\mathbf{n}$
 - **Normal stress:** Component σ_n of $\mathbf{t}$ parallel to $\mathbf{n}$
 - **Shear stress:** Component τ_n of $\mathbf{t}$ perpendicular to $\mathbf{n}$
 - Shear stress is an important cause of material damage
 - Plastic deformation: (Partly) irreversible when stress is removed



Invariants of Stress Tensors

- **Eigenvalues** of stress tensors are called **principal stresses**
 - Act on planes given by **principal directions**, i.e., the corresponding eigenvectors. For these, there is no shear stress.
 - Positive evals indicate *tensile* stress, negative evals *compressive* stress

- The **maximum shear stress** is given by

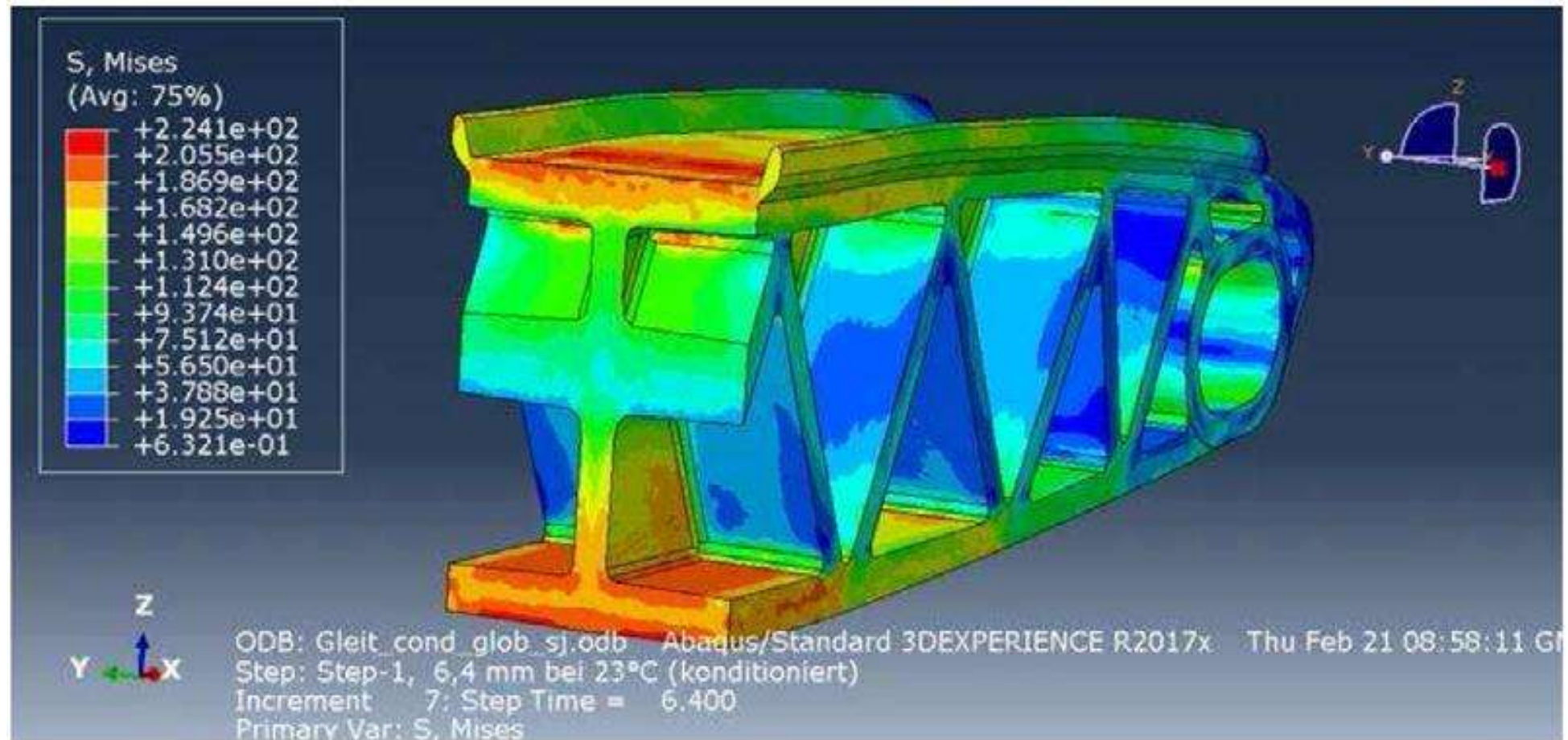
$$\tau = \frac{\lambda_1 - \lambda_3}{2}$$

- The **von Mises stress** determines the yield point of ductile materials

$$\sigma_v = \sqrt{\frac{(\lambda_1 - \lambda_2)^2 + (\lambda_2 - \lambda_3)^2 + (\lambda_1 - \lambda_3)^2}{2}}$$

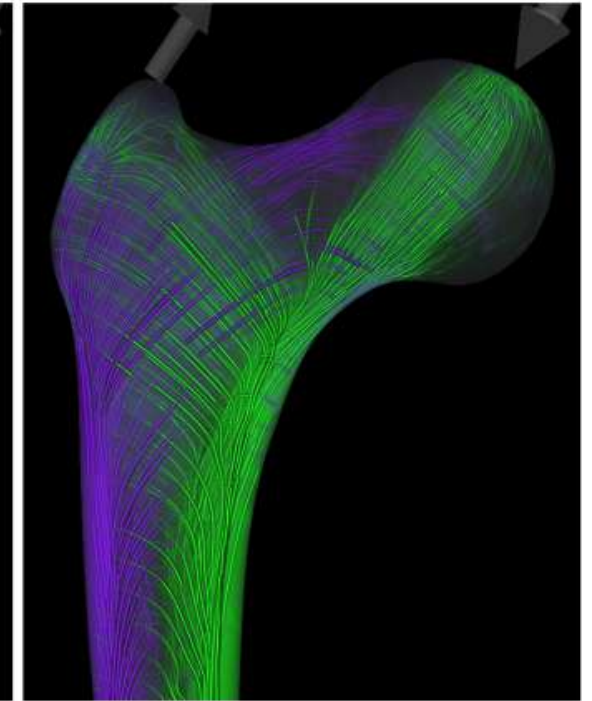
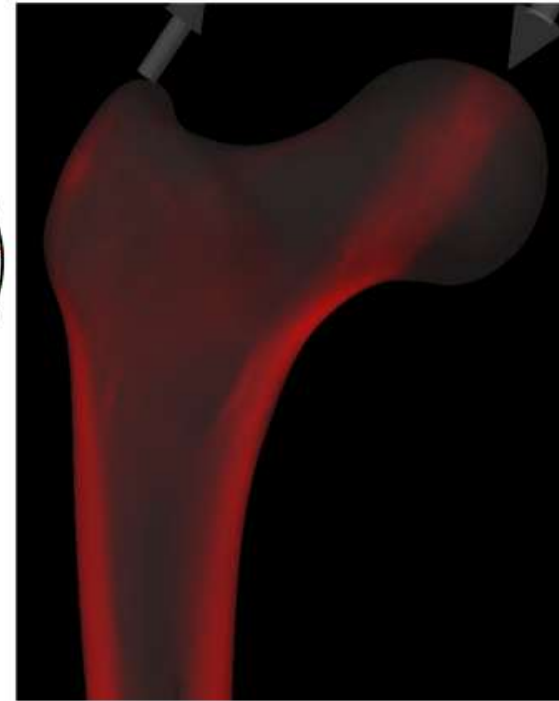
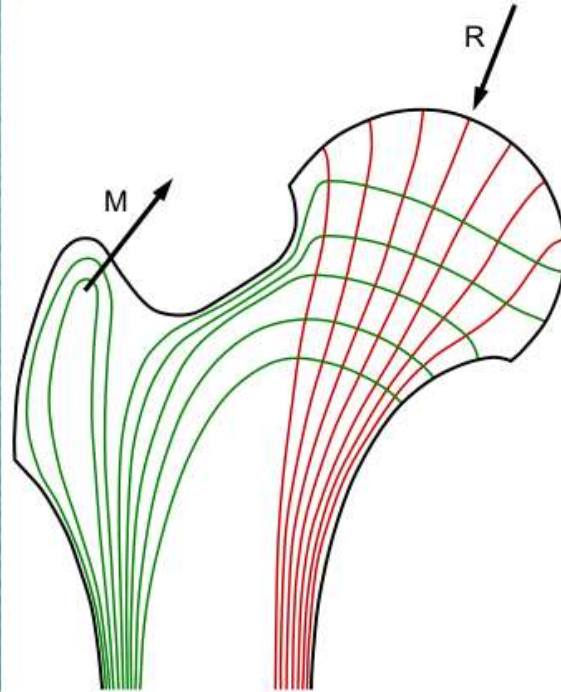
Examples: Visualizing Invariants of Stress Tensors

Example from [Hergl et al. 2021]: Color coding of von Mises stress on a CAD model



Examples: Visualizing Invariants of Stress Tensors

Example from [Dick et al. 2009]: Stress tensors in orthopedics



Trabecular structures within the bone (left) are aligned along principal stress directions (right)

Volume rendering of von Mises stress

Principal stresses (violet: tensile, green: compressive)

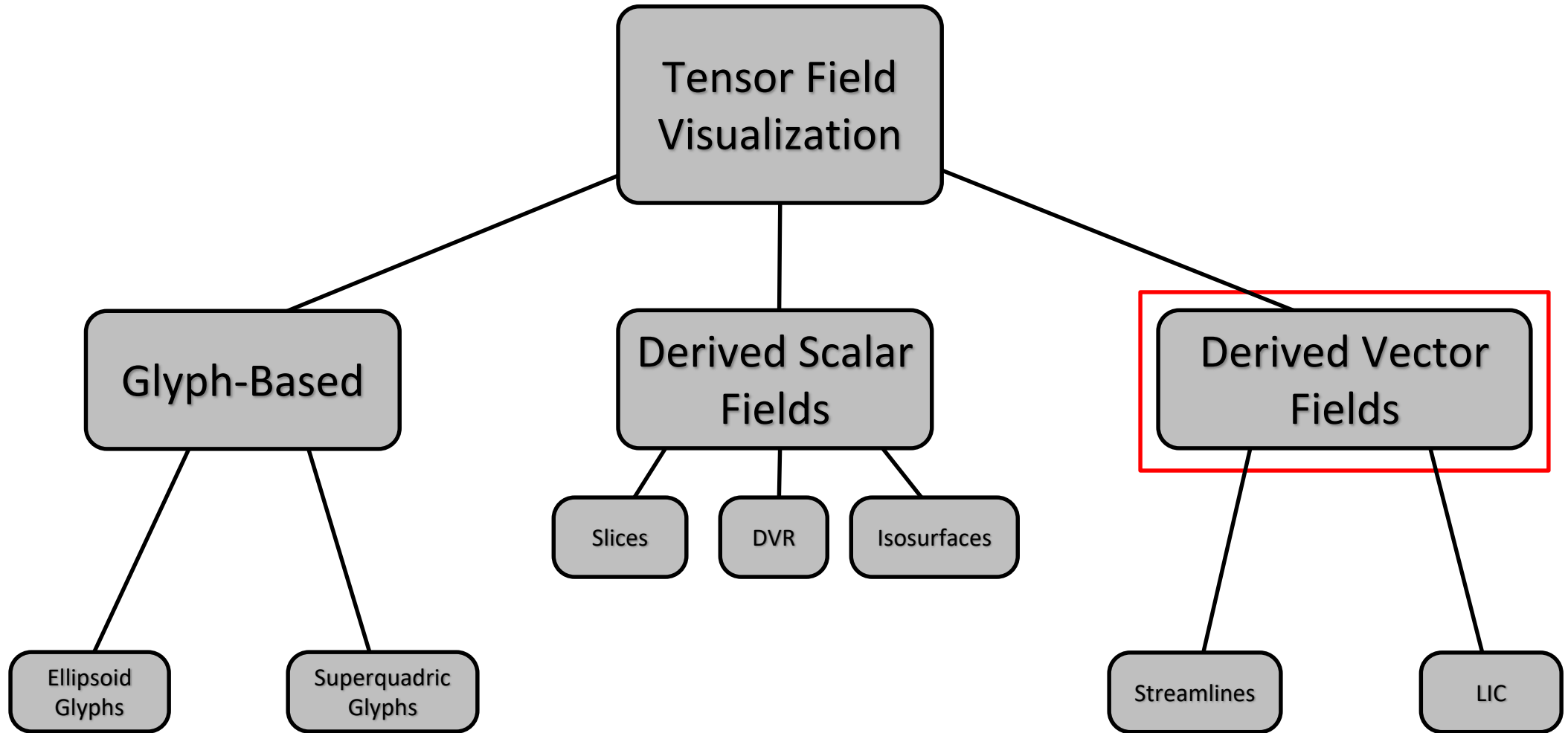
Summary: Derived Scalar Measures

- **Scalar invariants** are independent of the chosen frame of reference
 - Can be expressed in terms of eigenvalues
 - Can be used to apply standard volume visualization to tensors (e.g., isosurfaces, volume rendering)
- **In diffusion MRI:**
 - Overall diffusivity (MD), amount of anisotropy (e.g., FA), type of anisotropy (e.g., mode or Westin's measures)
- **In stress tensor fields:**
 - Principal stresses, maximum shear stress, von Mises stress

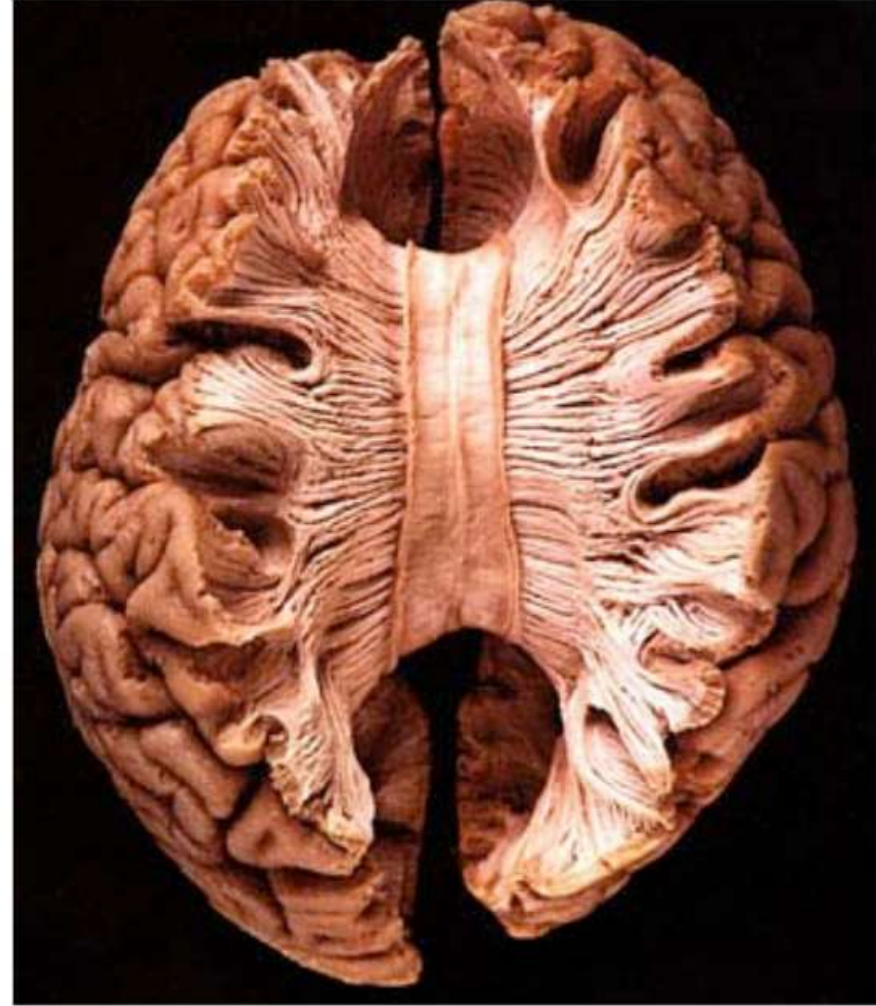
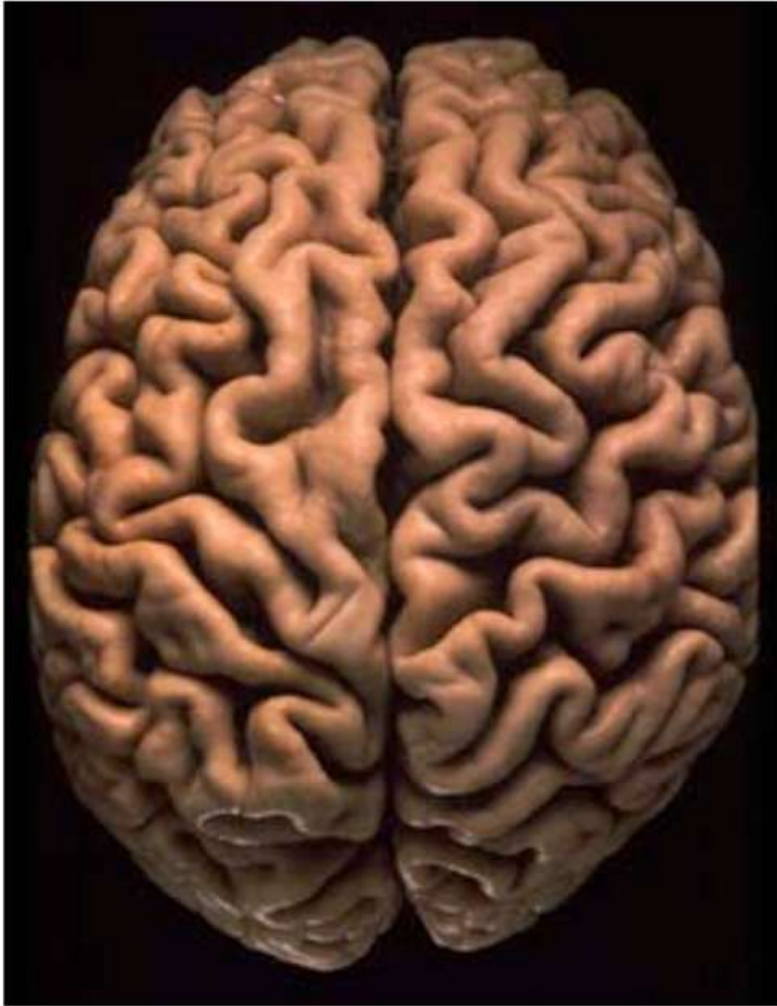
Section 14.4:

Visualizing Derived Vector Fields

Overview of Tensor Field Visualization



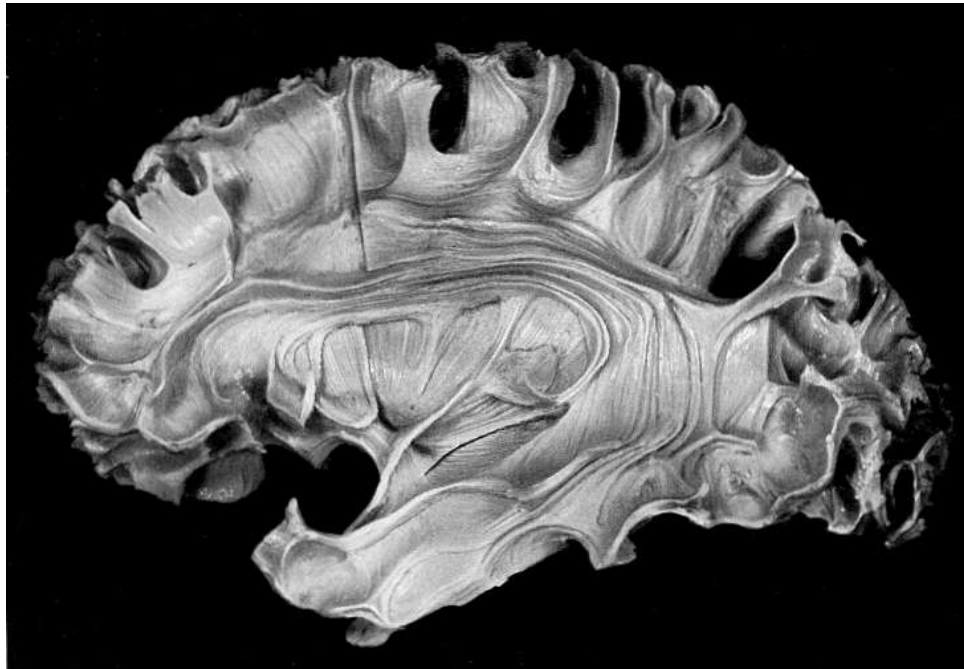
Fiber Bundles in the Brain



Klingler Dissection Method

- In anatomical textbooks: Klingler dissection
 - Developed by Josef Klingler in the 1930s
 - Freezing the brain and scratching away tissue
 - Reveals fiber tracts in their natural context

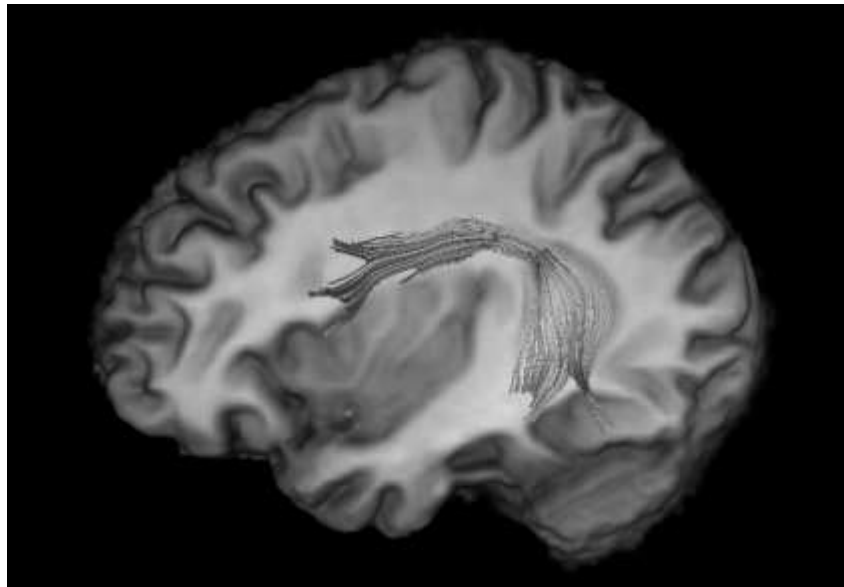
© S. Karger AG, Basel



Photograph of a
Klingler Dissection

Virtual Klingler Dissection

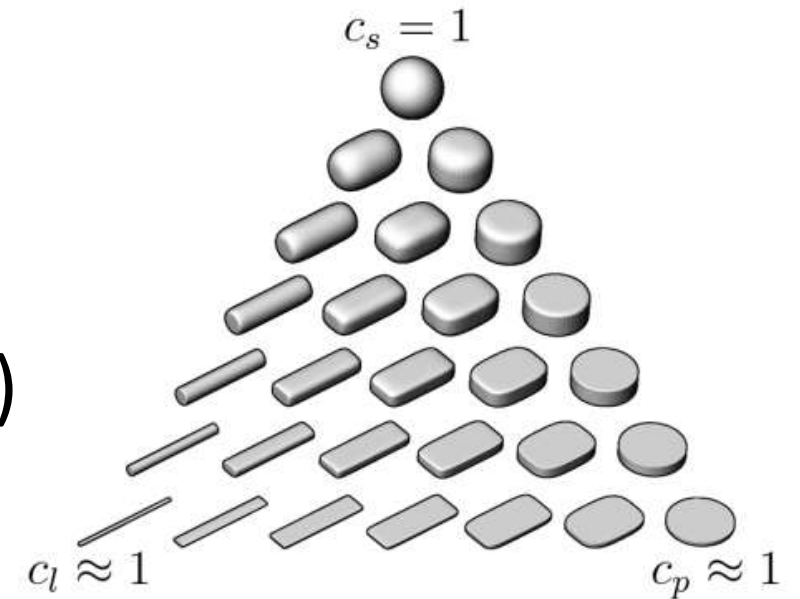
- Emulated Klingler Dissection with DT-MRI [Schultz et al. 2008]:
 1. **Reconstruct** tract of interest
 2. **Automatically cut away** a part of the brain
 3. **Deform the cutting plane** (“scratch away tissue”)
 4. **Render plane and fibers** with distance cueing



Example Result from a
Virtual Klingler Dissection

Visualizing Eigenvector Fields

- Most methods for vector field visualization can be applied to eigenvector fields
- **Constraint 1:** Unlike proper vectors, eigenvectors (i.e., $\mathbf{v}$ such that $\mathbf{M}\mathbf{v}=\lambda\mathbf{v}$) do not have an intrinsic norm or orientation
 - any scaled version $\alpha\mathbf{v}$ ($\alpha\neq 0$) is also an eigenvector
 - in particular, $-\mathbf{v}$ is also an eigenvector
- **Constraint 2:** When two (or more) eigenvalues are equal, $\mathbf{v}$ can be rotated freely in the corresponding two- (or higher) dimensional eigenspace

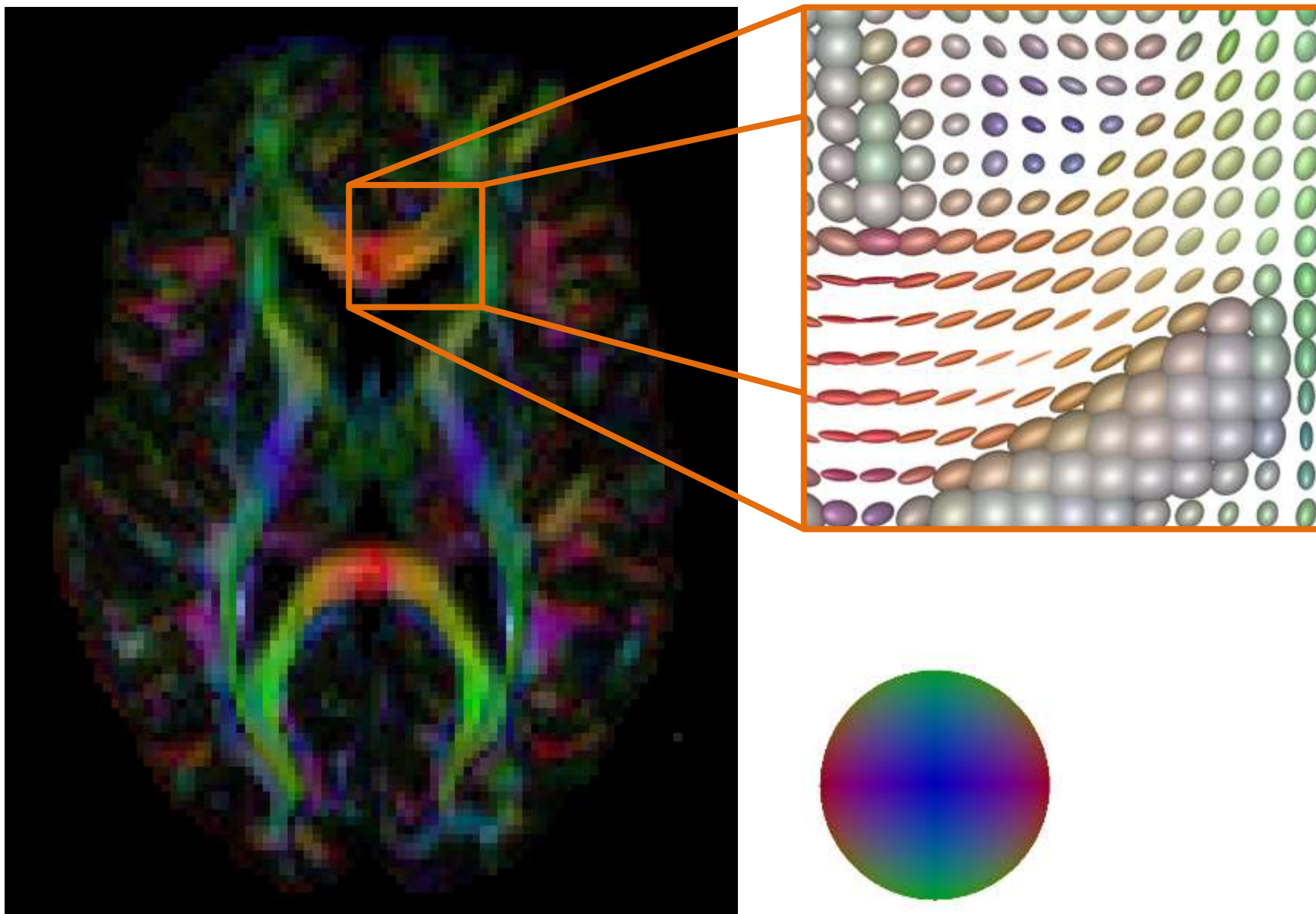


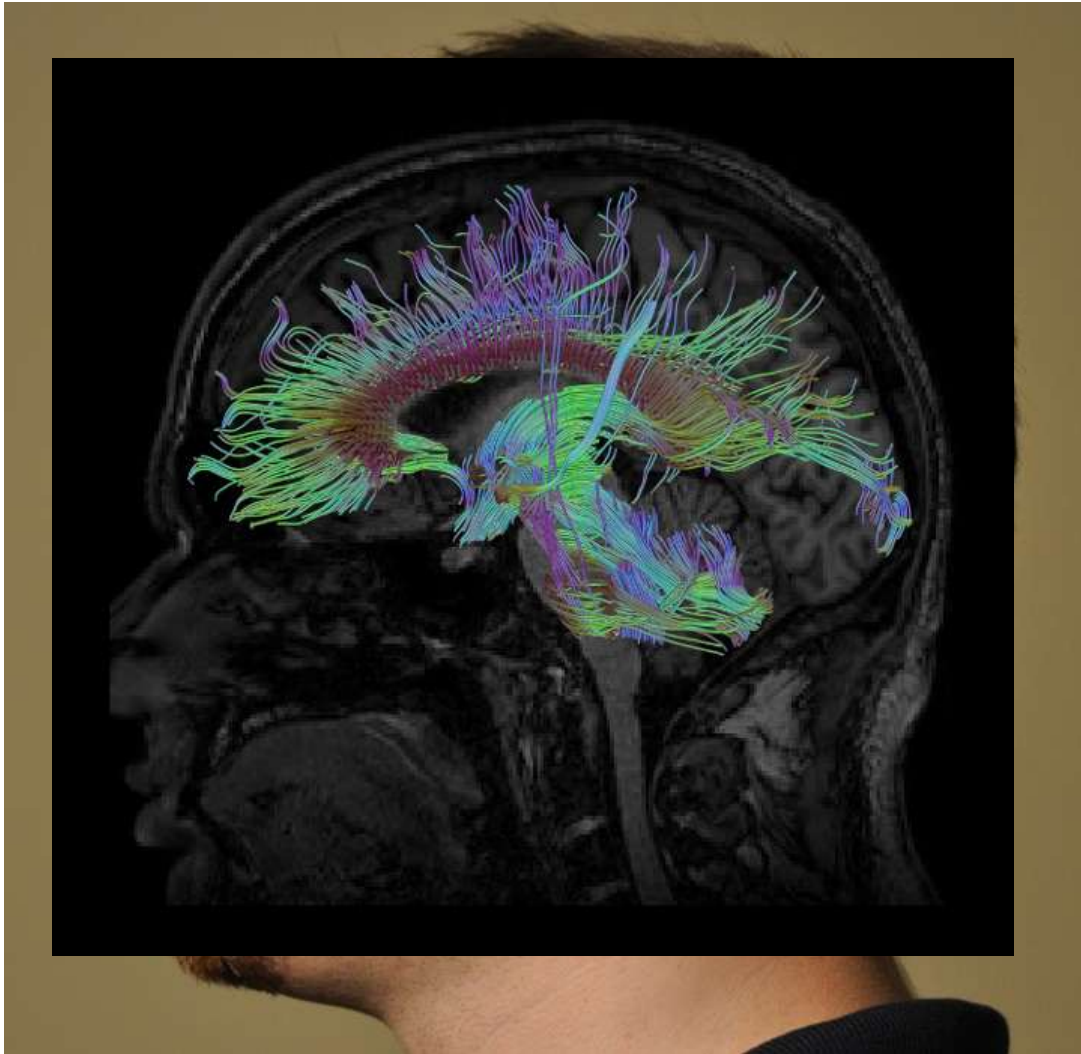
Fiber Tracking / Tractography

- In diffusion MRI, streamline integration in the principal eigenvector field $\mathbf{v}$ is called “fiber tracking” or “tractography”

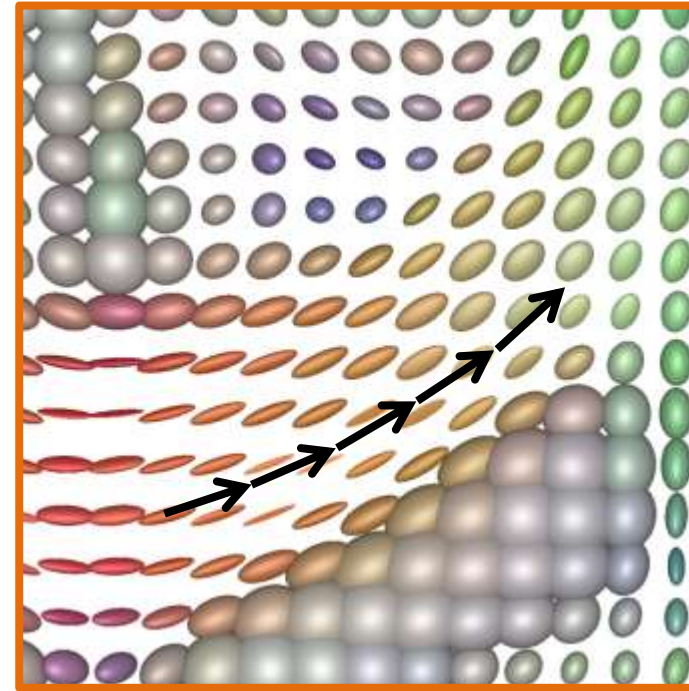
$$\frac{d}{d\tau} \mathbf{x}(\tau) = \mathbf{v}(\mathbf{x}(\tau))$$

- Numerical techniques from Section 11.4 can be used
 - Interpolation can be done by interpolating tensors, then computing $\mathbf{v}$
 - Sign of $\mathbf{v}$ can be selected so that dot product with previous step is >0
- Need additional *stopping criteria*
 - **Anisotropy:** Stop when FA or c_l are low (better to use c_l)
 - **Curvature:** Stop when dot product with previous vector is too small, or sum of angles over k previous steps is too large
 - **White matter mask:** Stop when leaving the white matter



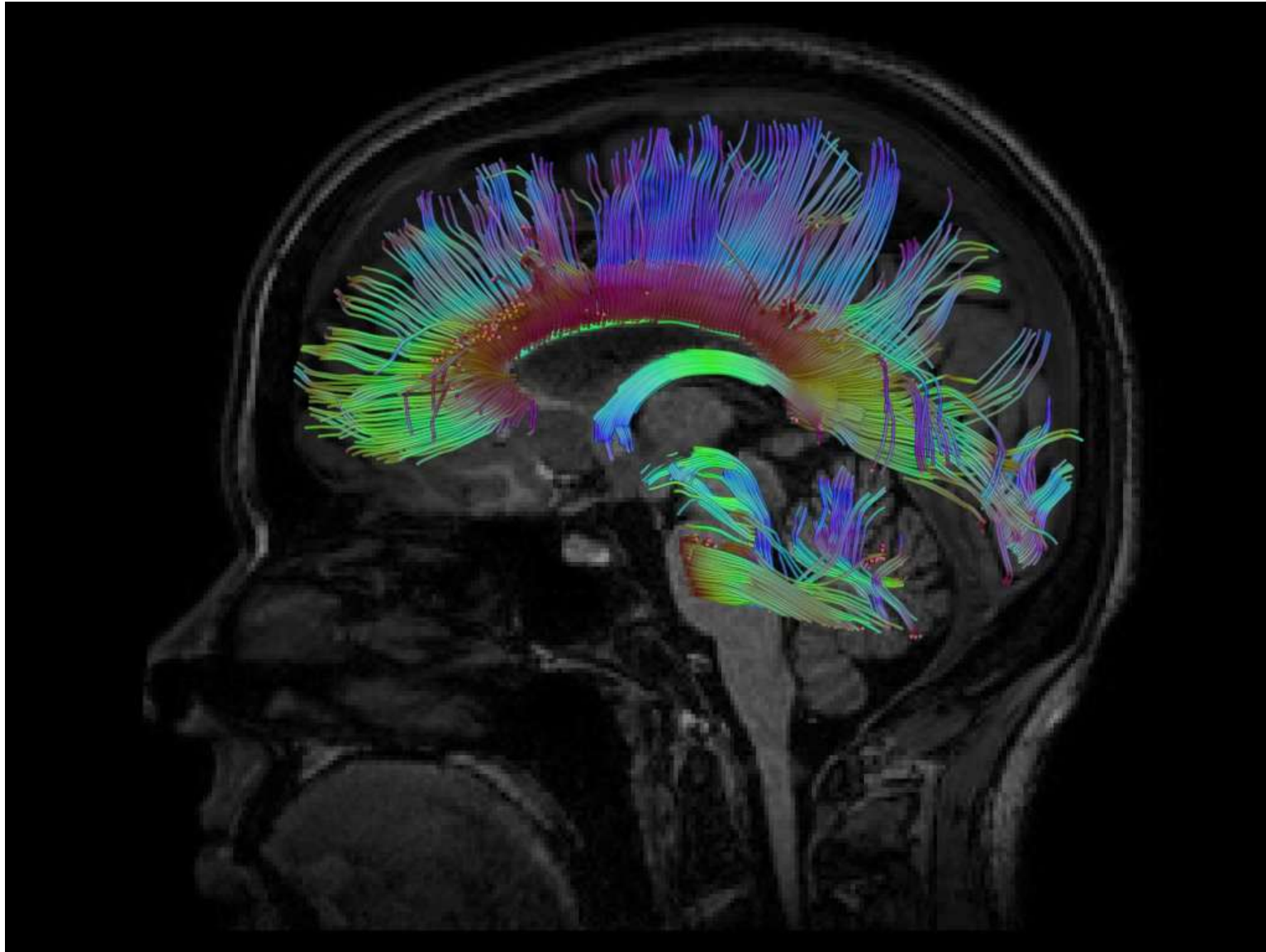


Seed Points on
Mid-Sagittal Plane



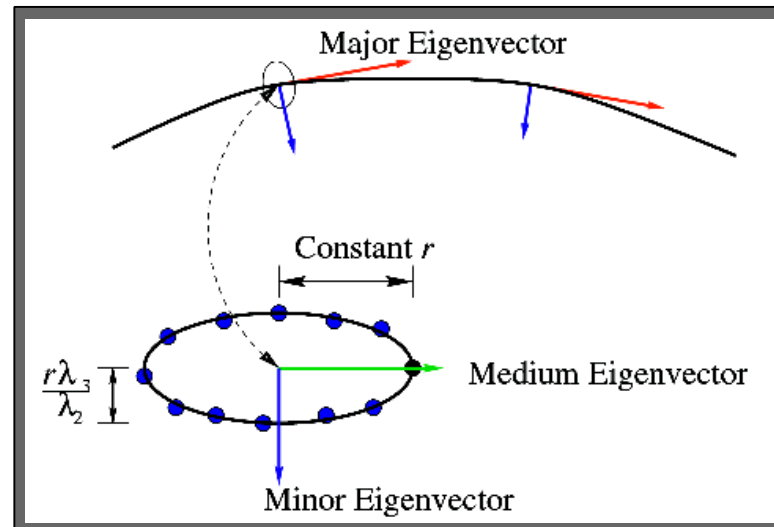
Deterministic
Tractography

Animated Tractography



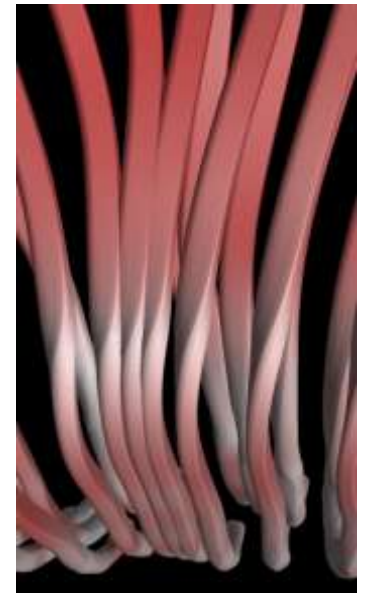
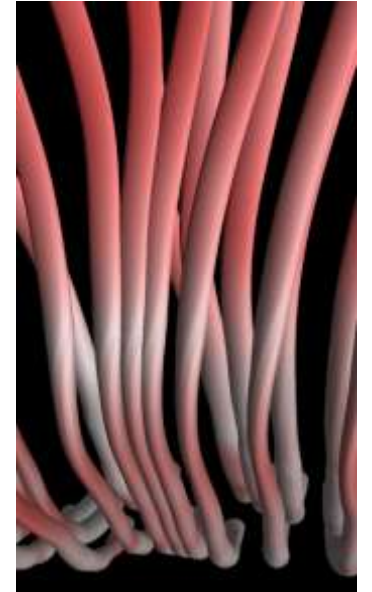
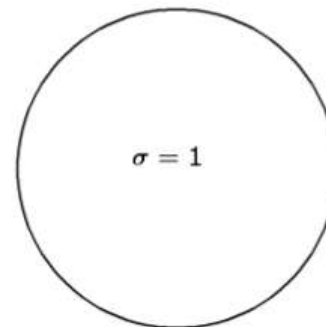
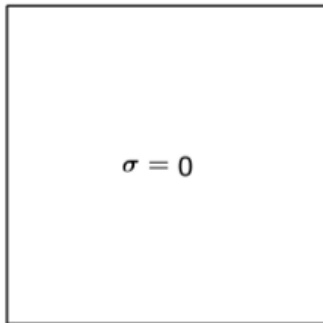
Streamtubes

- Stream tubes [Zhang et al. 2003] also encode second and third eigenvector
 - Elliptical cross-section reflects second/third eval
 - Fix maximum radius, preserve aspect ratio
 - Color indicates c_i (large c_i red)

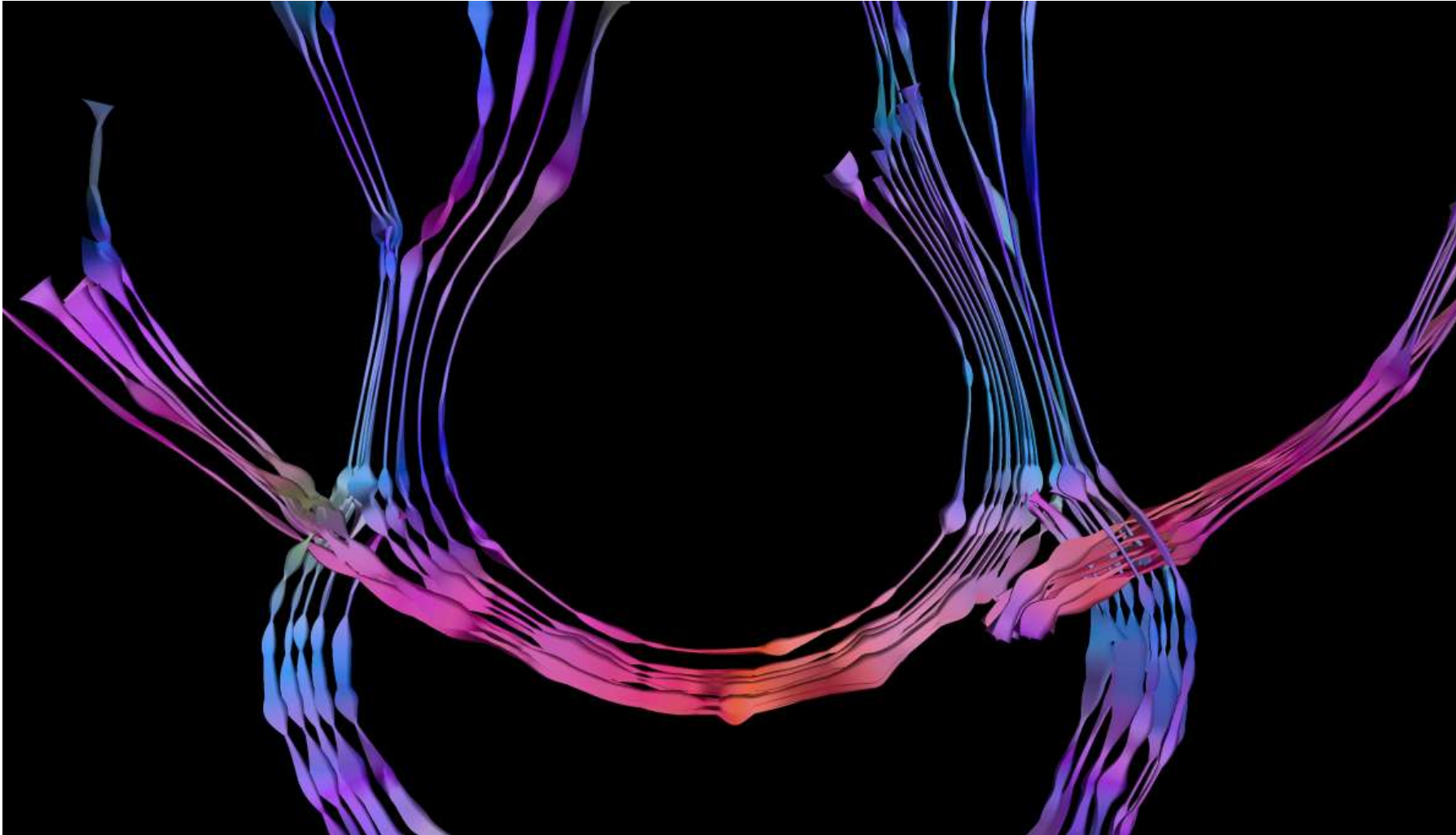


Superquadric Streamtubes

- Superquadric streamtubes [Wiens et al. 2014]
 - Superquadric instead of elliptical cross-section
 - Shape index $\sigma = \left(\frac{\lambda_3}{\lambda_2}\right)^\gamma$
 - Spherical for $\lambda_2 = \lambda_3$
 - Clear edges for $\lambda_2 \gg \lambda_3$

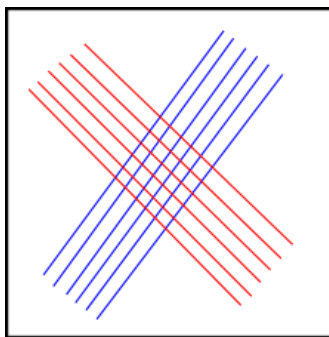
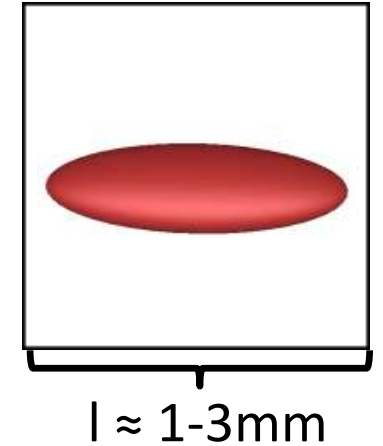


Superquadric Streamtubes: Example

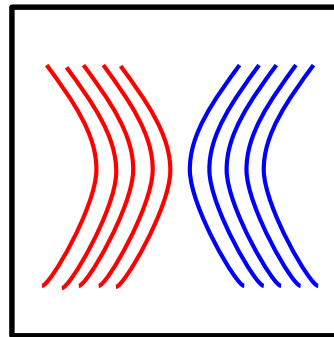


Caveats: Interpreting DT-MRI “Fibers”

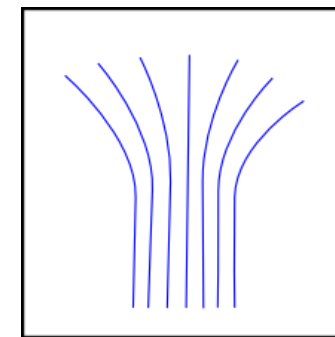
- Individual axons are much smaller than voxel size
 - axons often run in parallel (“fiber bundle”), making it possible to detect them at voxel level
 - Streamlines follow trajectory of fiber bundles
- Most voxels contain more complex fiber configurations
 - Details outside our scope, part of “Image Acquisition and Analysis in Neuroscience” lecture (planned in WiSe 25/26)



Crossing Fibers



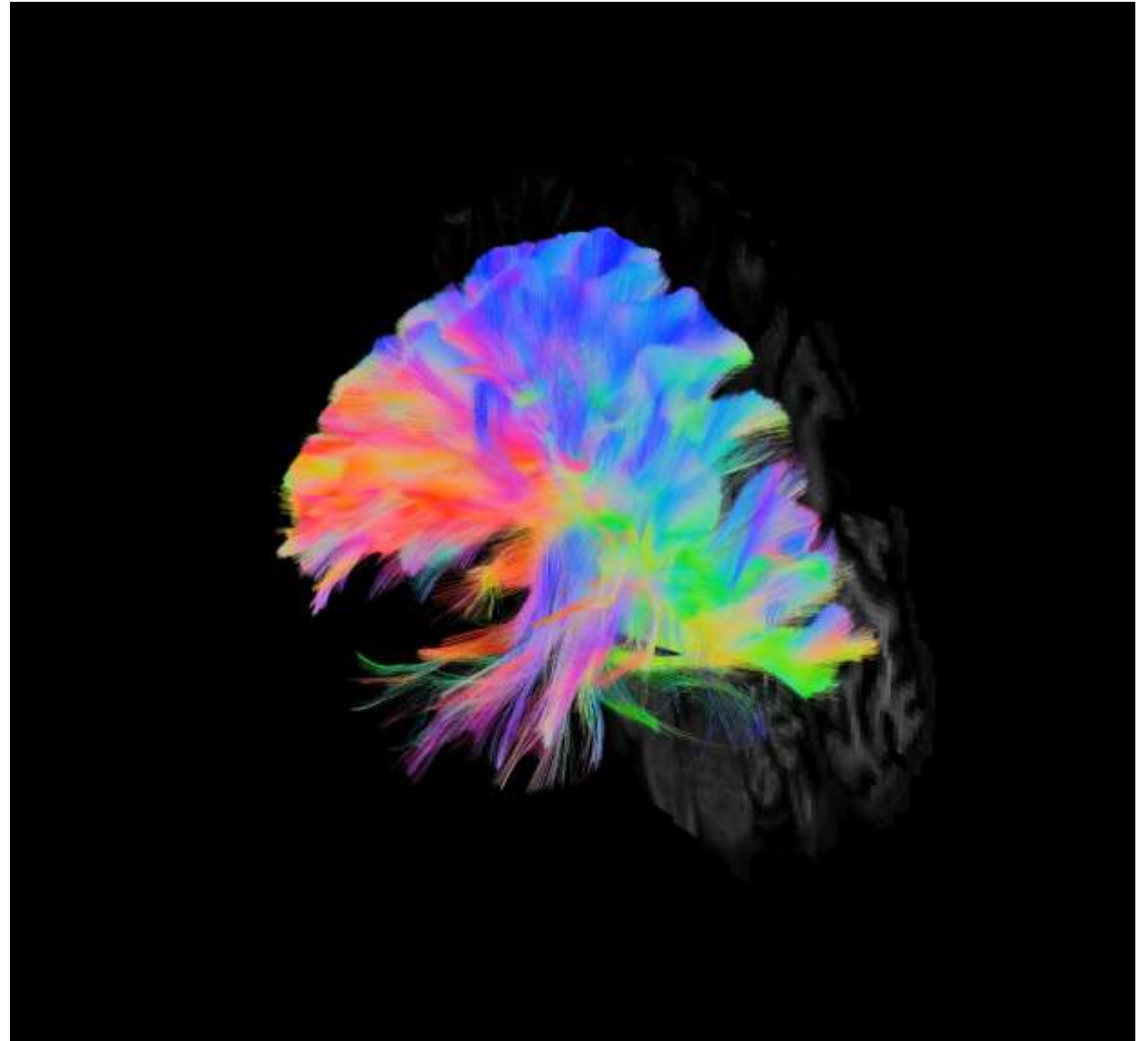
“Kissing” Fibers



Diverging Fibers

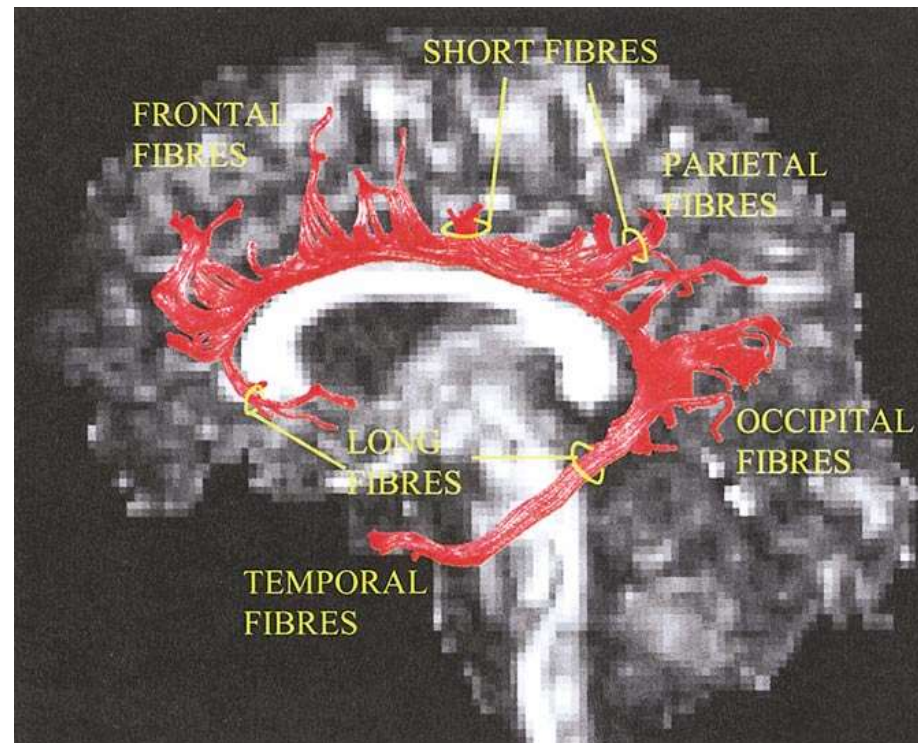
Example: Complex Fiber Configurations

Unscented Kalman Filter based
multi-fiber tractography of
corpus callosum, seeded at
mid-sagittal plane
[Gruen et al. 2023]



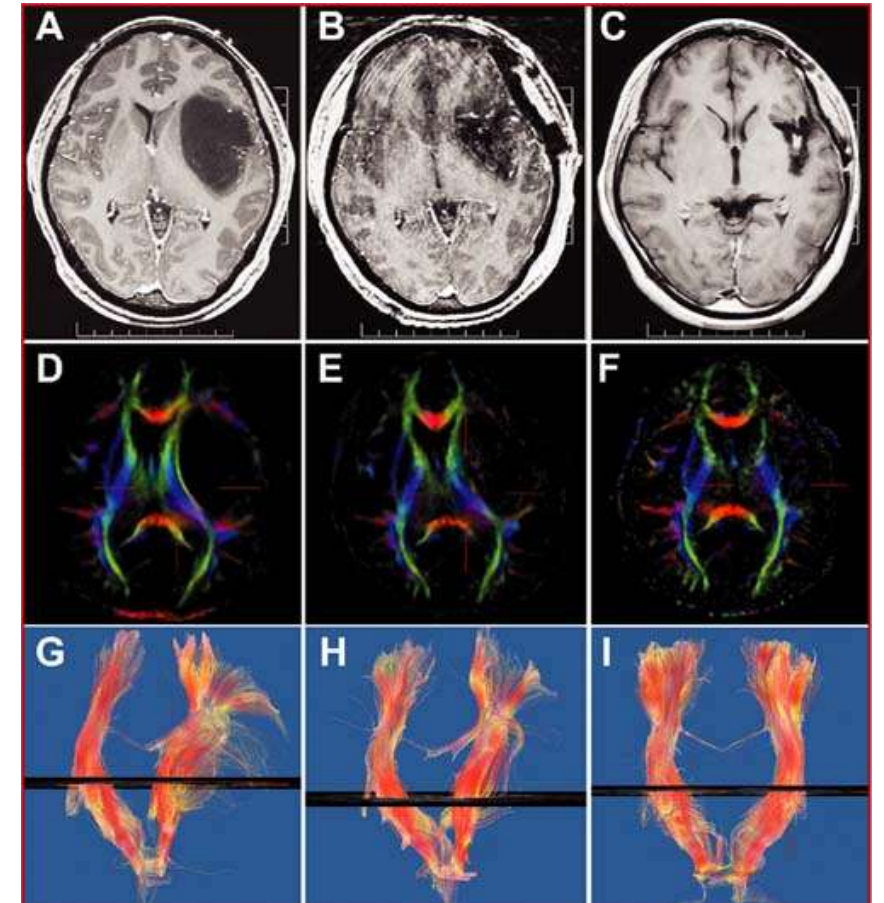
Success: White Matter Atlases

- **[Catani et al. 2002]** Deterministic tractography reliably reconstructs many previously known large-scale white matter tracts
 - Previously not possible to study them *in vivo*



Success: Surgical Planning

- **[Wu et al. 2007]** randomly either used (n=118) or did not use (n=120) dMRI for planning brain tumor surgery. They report that dMRI led to
 - Greater chance of complete resection
 - Reduced chance of postoperative motor deterioration
 - Longer survival
- **But:** Due to software constraints, mostly based on FA maps, fiber tracking only used in 23 cases

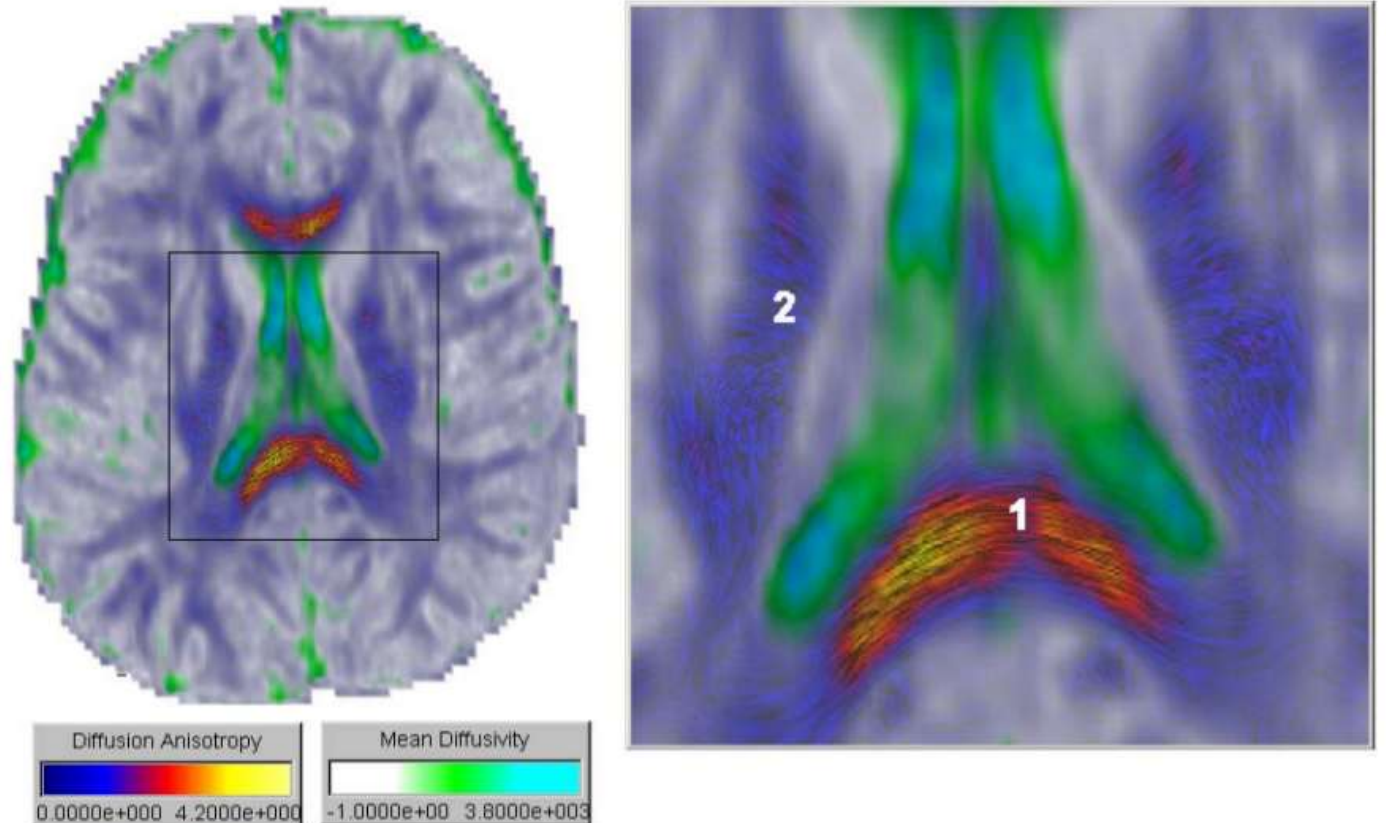


LIC on Eigenvector Fields

LIC generalizes in a straightforward way to eigenvector fields (account for orientation!)

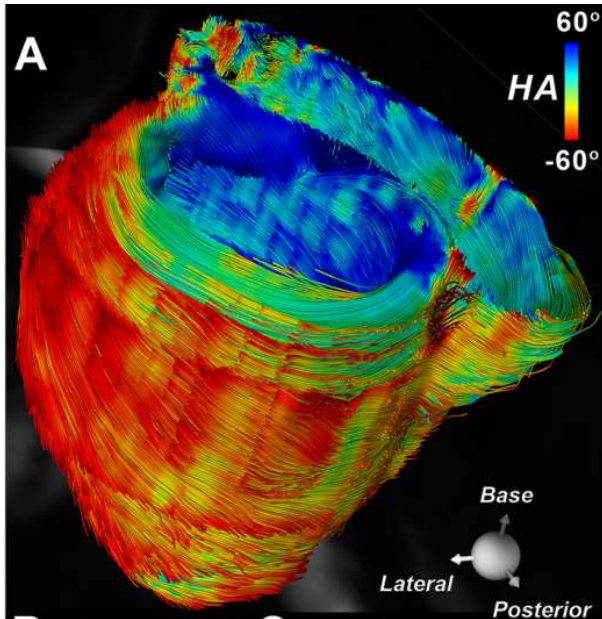
[Wünsche 2004]:

- Opacity of LIC texture depends on anisotropy
- Kernel length depends on angle with image plane
- Color code FA/MD

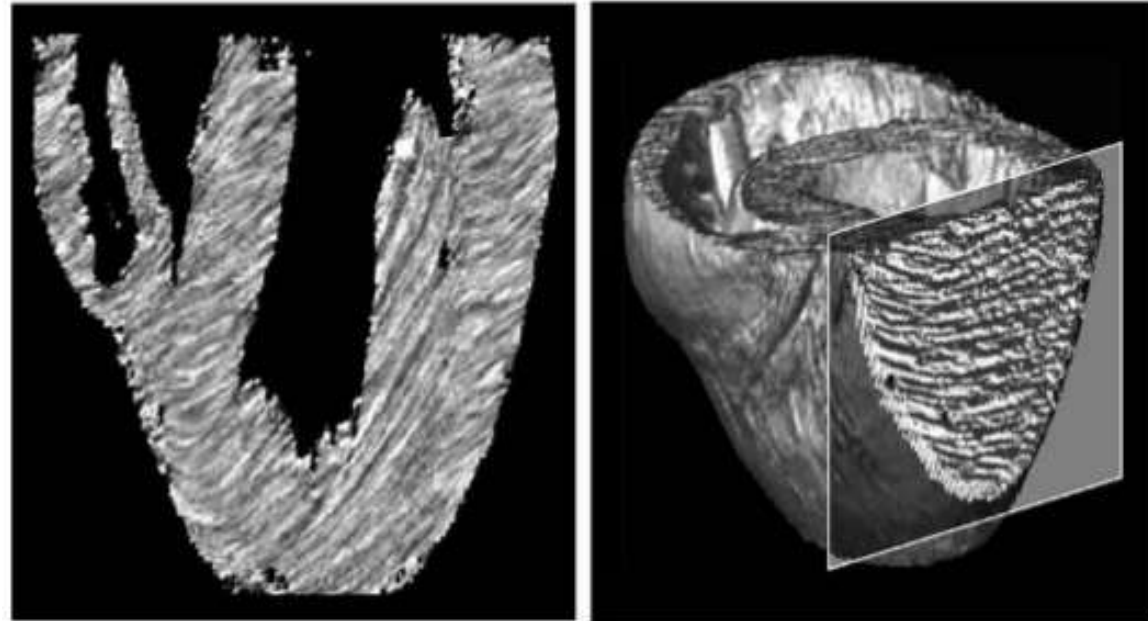


LIC of Myocardial DT-MRI

[Hsu 2001]: two-pass method that smoothes along two eigenvector fields



[Mekkaoui et al. 2015]

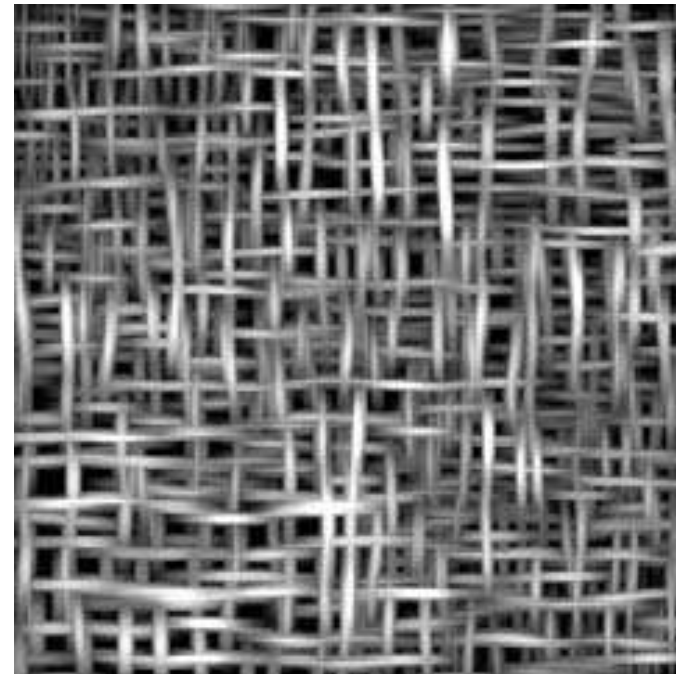
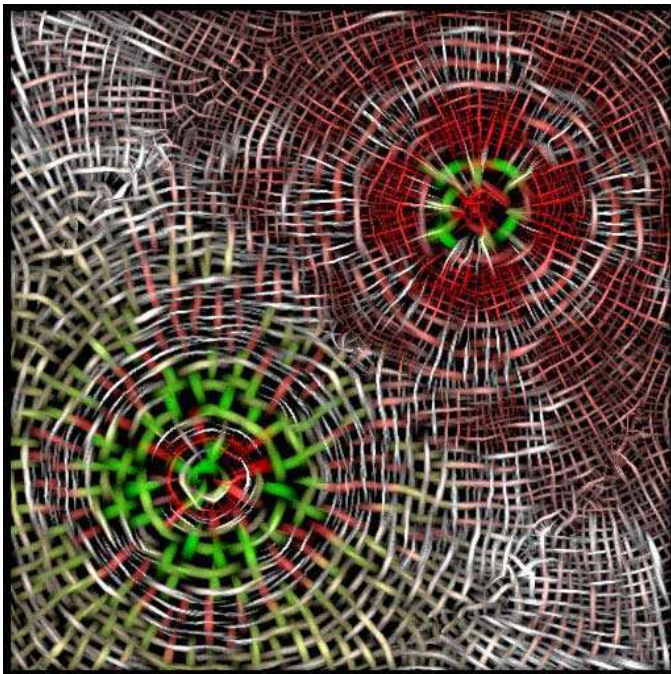


LIC of myocardial architecture

Fabric-like Textures

[Hotz et al. 2004]: Rather than iteratively smoothing the same texture along different eigenvector fields, create and average separate LIC images to create fabric-like appearance

- *Example:* Stresses in a two point load simulation



Summary: Vector-based Tensor Vis

- Most **vector field visualization** techniques can be applied to eigenvector fields
 - Streamline integration
 - Stream tubes encode additional eigenvectors
 - Line integral convolution
 - Account for lack of norm and orientation!
- In the context of diffusion MRI, **fiber tracking** traces curves that are tangential to inferred fiber direction
- In the context of **stress tensor fields**, streamlines illustrate principal stress directions

References

On Eigenvalues / Singular Values:

- Lloyd N. Trefethen, David Bau III: Numerical Linear Algebra, SIAM, 1997

On Diffusion Tensor Visualization:

- Thomas Schultz, Anna Vilanova: Diffusion MRI Visualization. NMR in Biomedicine 32:e3902, 2019
- Derek K. Jones: Diffusion MRI: Theory, Method, and Applications. Oxford University Press 2011

References: Tensor Glyphs

- Pierpaoli and Basser: “Toward A Quantitative Assessment of Diffusion Anisotropy”, Magnetic Resonance in Medicine 36: 893-906, 1996
- Kindlmann: “Superquadric tensor glyphs”, EG/IEEE Symposium on Visualization (SymVis), pages 147-154, 2004
- Kindlmann and Westin: “Diffusion Tensor Visualization with Glyph Packing”, IEEE Transactions on Visualization and Computer Graphics 12(5):1329-1335, 2006
- Schultz and Kindlmann: “Superquadric Glyphs for Symmetric Second-Order Tensors”, IEEE Trans. on Visualization and Computer Graphics 16(6):1595-1604

References: Deterministic Tractography

- Bassler, Pajevic, et al.: “In Vivo Fiber Tractography Using DT-MRI Data”, Magnetic Resonance in Medicine 44:625-632, 2000
- Mori, Crain, et al.: “Three-dimensional tracking of axonal projections in the brain by magnetic resonance imaging”, Annals of Neurology 45(2):265-269, 1999
- Schultz, Sauber, et al.: „Virtual Klingler Dissection: Putting Fibers into Context”, Computer Graphics Forum 31(3):1063-1070, 2008

Exam-like Questions

1. Name an application in which tensor fields arise, and state whether those tensors are (semi-)definite.
2. Which limitation of ellipsoid glyphs is resolved by superquadric glyphs and how?
3. Name a tensor invariant and explain its interpretation.
4. Propose a way in which Line Integral Convolution (LIC) can be adapted for tensor field visualization.

RWTH Teaching Evaluation

- RWTH Students can participate in the teaching evaluation at <https://evasys.rwth-aachen.de/evasys/online.php?pswd=RFHD7>
- **Open until July 16**
- *Note:* There are usually so few RWTH students that their feedback is not shared with me (for privacy reasons)
 - RWTH students are also welcome to participate in the lecture evaluation that will be held in Bonn, details will be shared later

